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Long-chain α-olefins production over Co-MnO_x catalyst with optimized interface

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A B S T R A C T

In this work, we report a carbon-mediated strategy to construct supported Co-MnO_x nano-interfacial catalyst that enables highly selective synthesis of long-chain linear α-olefins ($C_{5+}^{=}$) via the Fischer-Tropsch (FT) reaction. The co-impregnation of glucose as the carbon precursor and metal salt aqueous solution over SiO_2 followed by pyrolysis and oxidation creates a catalyst with stable and rich Co-MnO_x nano-interfaces after reduction and under FT conditions, which are convincingly revealed from characterizations of the catalysts at different stages. The optimal catalyst is stable for over 400 h without an observably decreased CO conversion, and the highest space-time yield of $C_{5+}^{=}$ is achieved to be 0.79 $g \cdot g_{Co}^{-1} \cdot h^{-1}$, which are mainly originated from the suppressed sintering of metallic Co and favored CO adsorption/dissociation with a reasonably inhibited H_2 co-adsorption. The simple procedure and broad availability of precursors make our developed strategy a practically important surface-engineering method capable of precisely fabricating stable nano-interfacial catalysts.

1. Introduction

The long-chain α-olefins ($C_{5+}^{=}$) are important starting chemicals for the industrial production of synthetic lubricant, plastic, detergent, and surfactant [1]. In comparison with the petrochemical route, Fischer-Tropsch synthesis (FTS) provides an attractive alternative since it can efficiently convert the syngas (H_2 +CO) to hydrocarbons (e.g., paraffins and olefins) with largely varied carbon numbers [2–4]. To date, great efforts with significant progresses have been made in regulating the product distribution of FTS either over Fe- or Co-based catalysts [5–8]. Importantly, lower olefins ($C_2^{=}$-$C_4^{=}$) with a selectivity as high as approximately 60% can be achieved over Al_2O_3-supported iron catalysts promoted with sulfur and sodium [9], or cobalt carbide (Co_2C) nanoprisms [10]. However, selective synthesis of $C_{5+}^{=}$ at a considerable activity level with a reasonably low CO_2 and CH_4 selectivity still remains a great challenge [11–13].

Co-based FTS catalysts are characterized by their excellent intrinsic activity, high selectivity of long-chain hydrocarbons, and low activity for water-gas shift reaction (WGS). Because of the high chain-growth probability and hydrogenation ability, Co-based catalysts mainly produce straight-chain paraffin products [14,15]. To maximize the catalytic activity and tune the product distribution, the introduction of promoters has been identified as an effective approach by modulating the electronic structure and chemical property of cobalt besides changing supports [16,17]. Among the reported promoters, manganese oxides (MnO_x) have long been found to be effective in increasing the CO consumption rate and boosting the selectivity of lower olefins [18,19], provided that the amount, location, introducing mode of MnO_x, and the preparation method of the catalysts are optimized [20]. In this case, the structural and electronic effects between MnO_x and Co are applied to elucidate the promotional effect of MnO_x on the catalytic phenomena although the detailed promoting manner is far behind clear. Significantly, the interface between the intimately interacted Co and MnO_x is found to facilitate the dissociation & disproportionation of CO due

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generally to the Lewis acidity of MnO_x [21,22], while in turn hinders the competitive adsorption of H_2. Thus, the surface coverage ratio of adsorbed CO* (C*) to H* is increased around the interfaces of Co and MnO_x.

According to the polymerization kinetics of FTS, carbon chains are prolonged via the stepwise polymerization on the surface of metal catalyst. The termination of hydrocarbon chains may involve the hydrogenation of the adsorbed hydrocarbon fragments to form *n*-paraffins or the elimination of β-hydrogen to generate α-olefins. The primary α-olefins may further participate in the secondary reactions, e.g., hydrogenation & polymerization, to produce *n*-paraffins or reinsertion into chain-growth reactions [23,24]. Thus, the distribution of specific products is determined by the surface reaction rates of the chain growth and termination. In this case, rates of the carbon-chain growth and β-H elimination are expectedly enhanced by a higher surface coverage ratio of CO* (C*)/H* , leading to a higher selectivity of long-chain olefins [25]. Following this understanding, an interfacial catalsyst with plentiful Co-MnO_x nano-interfaces may promote the selectivity of long-chain olefins via increasing the surface CO* (C*)/H* ratio. Thus, different strategies have been practiced to regulate the interactions between Co and MnO_x via adjusting their spatial proximity, e.g., strong electrostatic adsorption [26], pretreatment technique [27], and support modification [28,29]. Unfortunately, the segregation and the agglomeration of MnO_x from metallic Co commonly occurs during the reduction of the catalyst and under the typical FTS conditions [30–32]. In principle, metallic Co with smaller size, creates more interfacial sites if MnO_x can be deposited in a controlled manner. Moreover, the intrinsic size effect of metallic Co on the FTS reaction is commonly observed [33,34]. Thus, the promotional effect of MnO_x on Co can be maximized if a stable Co-MnO_x nano-interface with a maximized perimeter is realized via tuning the Co size and Co-MnO_x interactions.

In this work, a carbon-mediated strategy is proposed to construct a robust nano-interfacial Co-MnO_x-(C)/SiO_2 catalyst (Scheme 1). Specifically, given the chelating property of glucose with metal ions [35], glucose and metal precursors are first co-impregnated on SiO_2 support, and the succeeded pyrolysis (carbonization) and calcination (oxidation) lead to the oxidized Co-MnO_x-(C)/SiO_2 catalyst. The resulting small Co_3O_4 particles are endowed with abundant surface oxygen vacancies (low-coordinated octahedral Co^{2+} ions), which facilitate dispersing and anchoring of MnO_x species. Stable and rich Co-MnO_x nano-interfaces between Co nano particles (NPs) and MnO_x islands are obtained after reduction and under FTS reaction conditions, leading to a high activity and long-term stability. The olefin selectivity of 56% with 66% $C_{5+}^{=}$ is achieved at 210 ºC over the optimal Co-MnO_x-(C)/SiO_2 catalyst with a CO conversion of 11.5%. Importantly, the catalyst achieves the highest space-time yield (STY) of $C_{5+}^{=}$, i.e., $STY_{Cat.} = 0.103\ g \cdot g_{cat.}^{-1} \cdot h^{-1}$ and $STY_{Co} = 0.79\ g \cdot g_{Co}^{-1} \cdot h^{-1}$ at 230 °C based on the mass of catalyst and metallic cobalt, respectively. The abundance and versatility of the carbon precursor and the facile impregnating process make the developed strategy a potentially important industrial method for fabricating effective and stable metal-oxide nano-interfacial catalyst.

2. Experimental section

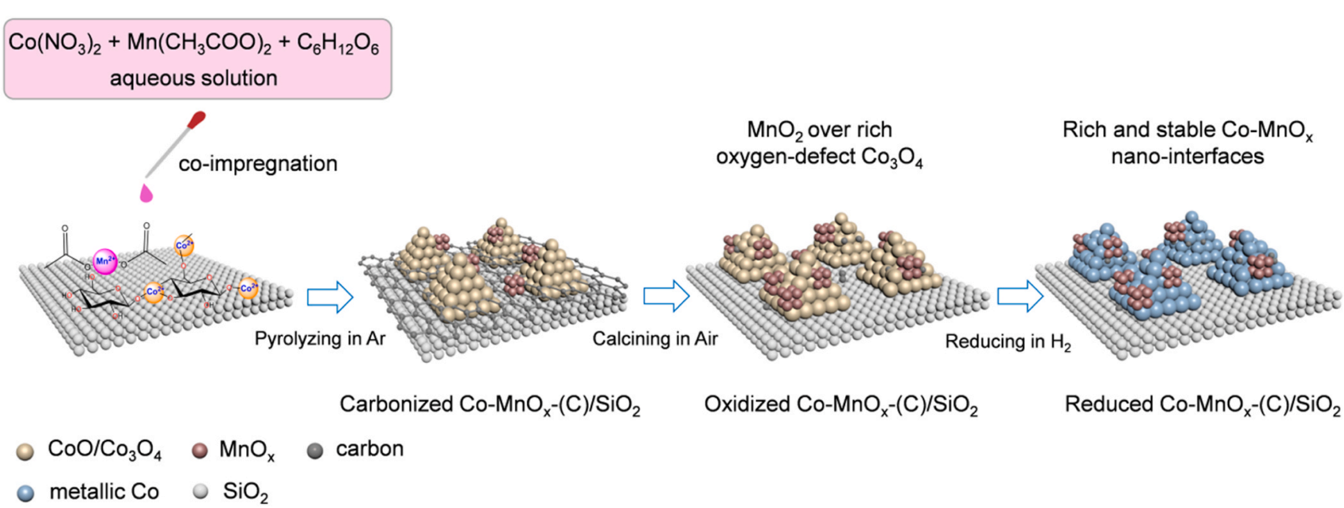
2.1. Materials

Cobalt(II) nitrate hexahydrate ($Co(NO_3)_2 \cdot 6\ H_2O$, ≥ 98%, CAS: 10026–22-9), Manganese(II) acetate tetrahydrate ($Mn(CH_3COO)_2 \cdot 4\ H_2O$, ≥ 99%, CAS: 6156–78-1) and glucose (≥99%, CAS: 50–99-7) were purchased from Shanghai Chemical Corp. and were used as received without any further purification. SiO_2 (Fujisilicia Q15) was purchased from Fuji Silicia Chemical Co., Ltd. and was pretreated by calcination in air at 600 °C for 3 h.

2.2. Synthesis of catalysts

The carbon-mediated strategy was accomplished in three steps, i.e., the co-impregnation of the aqueous solution of $Co(NO_3)_2$ as the cobalt precursor and glucose as the carbon source over SiO_2 followed by drying, the pyrolysis of the dried solid in an inert gas, and the subsequent oxidation in the air atmosphere. Specifically, the detailed procedure for synthesizing the Co-(C)/SiO_2 catalyst, where C stands for the carbon-mediated strategy, was explained as follows. Firstly, by adding 2.0 g SiO_2 powder in an aqueous solution of 1.48 g $Co(NO_3)_2 \cdot 6\ H_2O$ and 0.4 g glucose dissolved in 20 mL deionized water, the co-impregnation was performed at 60 °C for 90 min under a constant stirring of 400 round per minute (rpm). After removing the solvent in a rotary evaporator (100 rpm) at 60 °C, the solid was dried at 80 °C for 10 h. Subsequently, the dried sample was transferred to a tubular furnace, and the pyrolysis was conducted at 400 °C (5 °C/min) for 5 h in an Ar flow of 50 mL/min. Finally, the pyrolyzed sample was calcined in a furnace at 400 °C (5 °C/min) for 3 h in the static air atmosphere, leading to the Co-(C)/SiO_2 catalyst.

For the synthesis of Mn-promoted catalyst of Co-MnO_x-(C)/SiO_2 via the carbon-mediated strategy, the co-impregnation was performed by adding 2.0 g SiO_2 in an aqueous solution of 1.48 g $Co(NO_3)_2 \cdot 6\ H_2O$, 0.4 g glucose, and a stoichiometric ratio of $Mn(CH_3COO)_2 \cdot 4\ H_2O$ dissolved in 20 mL deionized water. $Mn(CH_3COO)_2 \cdot 4\ H_2O$ was added to



Scheme 1. The carbon-mediated strategy for the synthesis of Co-MnO_x-(C)/SiO_2 catalyst. The carbon-mediated strategy illustrated by using glucose as the carbon precursor for the synthesis of Co-MnO_x-(C)/SiO_2 catalyst via the consecutive steps of co-impregnation, pyrolysis, calcination, and reduction.

achieve molar Mn/Co ratios of 0, 0.02, 0.05, 0.1, 0.2, and 0.5. If no specific value is provided, the Mn/Co molar ratio is defaulted to 0.2. After the impregnation, exactly the same procedure and parameters to those of the Co-(C)/SiO_2 catalyst were applied, leading to Co-MnO_x-(C)/SiO_2.

For a comparison purpose, the reference catalysts were synthesized without adding glucose as the carbon source during the impregnation step. Specifically, 2.0 g SiO_2 powder was added into the aqueous solution of 1.48 g $Co(NO_3)_2\cdot6\,H_2O$ dissolved in 20 mL deionized water to implement the impregnation for synthesizing Co/SiO_2. In the same way, 2.0 g SiO_2 added in the aqueous solution of 1.48 g $Co(NO_3)_2\cdot6\,H_2O$ and 0.25 g $Mn(CH_3COO)_2\cdot4\,H_2O$ dissolved in 20 mL deionized water was used for the synthesis of Co-MnO_x/SiO_2. To minimize any potential effects, the steps of the impregnation, drying, pyrolysis, and calcination together with the parameters used for the carbon-mediated strategy were followed and kept the same for synthesizing Co/SiO_2 and Co-MnO_x/SiO_2 catalysts.

2.3. Material characterizations

The metal loadings were analyzed by using inductively coupled plasma-optical emission spectroscopy (ICP-OES, SPECTRO ARCOS). N_2 adsorption-desorption isotherms were tested at $-196\,^\circ C$ using a Belsorp-Max (Bel Japan Inc.) apparatus. Prior to measurement, the samples were degassed under vacuum at 300 °C for 10 h. X-ray diffraction (XRD) patterns were obtained on a Bruker D8 Advance X-ray diffractometer with monochromatic Cu Kα radiation (λ = 0.15406 nm) at 40 kV and 40 mA with a scanning speed of $0.5^\circ\cdot min^{-1}$. The in situ XRD experiments were performed at the BL14B1 beamline ($\lambda = 0.6887$ Å, 10 keV) of the Shanghai Synchrotron Radiation Facility. The catalyst in the in situ cell was heated from 40 to 400 °C (5 °C/min) with a flow of 10% H_2/Ar at atmospheric pressure. Raman spectra were collected on a Raman spectrometer (LABRAM HR EVO, Horiba) with the excitation source laser beam of 532 nm. The wavenumber was calibrated by the peak at 520.7 cm^{-1} of a silica standard. High-resolution transmission electron microscopy (HR-TEM) images and high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) together with the elemental mapping were recorded on a TEM (FEI, TecnaiG2 F20) equipped with a Gatan Tridiem 863 P post column image filter (GIF) and a high angle energy dispersive X-ray (EDX) detector operating at 200 kV. The particle size was determined from TEM images and calculated upon the measurement of random 300 particles.

X-ray photoelectron spectroscopy (XPS) analyses were performed on an Axis Ultra spectrometer (Kratos Analytical Ltd.) using Al monochromatic X-ray source at room temperature in a high vacuum environment (approximately 5×10^{-9} torr). The binding energy was calibrated to the containment carbon C1s peak (284.8 eV). For the pseudo-in situ XPS study, the sample was made into small disk (6 mm diameter) and held on the sample holder. After pretreated in a flow of 10% H_2 in Ar (10 mL/min) at 400 °C for 3 h and followed by reacting in syngas (32%CO/64%H_2/4%Ar, 10 mL/min) at 220 °C for 10 h, the sample was transferred directly into an ultrahigh-vacuum chamber for XPS measurement without exposure to air. The spectra of samples after reduction or reaction were obtained under ultrahigh vacuum conditions using the monochromatic Al Kα radiation. X-ray adsorption spectra of Co K-edge were obtained on the 1W1B beamline at the Beijing Synchrotron Radiation Facility (BSRF, China). The storage ring energy was 2.5 GeV with an average current of 250 mA. Energy calibration of the Co and Mn K-edge were performed using Co and Mn foil, respectively. All samples were grinded into fine powders and brushed onto adhesive tapes. Data processing was conducted by using a standard procedure of background absorption removal and normalization.

H_2 temperature programmed reduction (H_2-TPR) and O_2 titration were performed on a Micromeritics ASAP 2920 apparatus. Reduction degrees of the reduced catalysts were determined by O_2-titration at 400 °C in a flow of Ar (30 mL/min). The amount of surface metallic Co atoms was measured by the H_2 chemisorption experiment on Micromeritics ASAP 2020 C apparatus. Prior to the measurement, 100 mg of sample was reduced in pure H_2 (50 mL/min) at 400 °C for 5 h. After purging and cooling, the measurement was performed at 150 °C. The cobalt dispersion (D%) was calculated according to the equation $D = 1.179X/(W\times f)$, where X is the total H_2 uptake in $umol\cdot g_{cat}^{-1}$, W is the weight percentage of cobalt, and f is the reduction degree calculated from O_2-titration. The particle size of cobalt was estimated from the H_2 uptake with the calibrated reduction degree, assuming an atomic ratio of HCo = 1 and a hemispherical crystallite geometry with a surface atomic density of 14.6 atoms/nm^2.

In situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) measurements were collected on a Bruker VERTEX70 spectrometer equipped with a MCT detector cooled by liquid N_2 and an in situ cell (Diffuse IR, PIKE Company, American) containing CaF_2 windows. For the in situ DRIFTS of CO adsorption, specifically, 50 mg of solid sample was loaded in the in situ cell and treated in a pure H_2 flow (20 mL/min) at 400 °C for 5 h under atmospheric pressure. After cooled down to room temperature, the sample was switched to a flow of Ar (20 mL/min) and the background spectrum was recorded. Then, the sample was switched to a flow of 30% CO in Ar (20 mL/min) and heated by stepwise increasing temperature from 40 to 220 °C (5 °C/min). The spectra for CO adsorption were collected every 10 °C at a resolution of 4 cm^{-1} and 64 scans. For the in situ DRIFTS experiment of CO hydrogenation, the reduced sample was flushed with Ar and the background spectrum was recorded. Afterwards, the sample was exposed to syngas (32%CO/64%H_2/4%Ar, 20 mL/min) and heated by stepwise increasing temperature from 40 to 230 °C under 0.6 MPa. The spectra were collected every 10 °C at a resolution of 4 cm^{-1} and 64 scans.

2.4. Catalytic performance test

The FTS reactions were performed in a high-pressure stainless-steel fixed-bed reactor with an inner diameter of 9 mm. Typically, 0.5 g catalyst (40–60 mesh) diluted with an equal volume of quartz sands (40–60 mesh) was loaded at the isothermal zone of the reactor. Before the FTS test, the catalyst was reduced at 400 °C and 0.1 MPa for 5 h in a pure H_2 flow of 50 $mL\cdot min^{-1}$. After the reduction, the reactor was cooled to 120 °C in the H_2 flow. Then, the syngas with the volumetric composition of 32% CO, 64% H_2, and 4% Ar as the internal standard was switched, and pressurized to 1.0 MPa. The catalytic test was conducted at the reactor temperature range from 210 to 230 °C and the gas hourly space velocity (GHSV) from 2.4 to 9.6 $L\cdot g_{cat}^{-1}\cdot h^{-1}$. The tail gas out of the reactor was depressurized before entering an online gas chromatograph (GC), and the entire pipeline was kept at 300 °C to avoid the product condensation. The gases of CO, CH_4, Ar, and CO_2 in the effluent were separated using PROPAK Q and CST-TDX-01 packed columns, and analyzed on a thermal conductivity detector (TCD). Hydrocarbons (C_1–C_{18}) were separated on an HP-PONA capillary column and analyzed using a flame ionization detector (FID). For all of the FTS experiments, the carbon balance was achieved at 100 ± 10%. An illustrative scheme of the reaction system, the detailed procedure for the FTS experiments, the analyses of the products, and the calculations are given in Section 1.1 of the Supplementary material.

3. Results and discussion

3.1. Catalytic performance

The nominal loading of cobalt over all of the catalysts is fixed at 13.0 wt%, and the Mn/Co molar ratio is set to be 0.2 if it is not specified, which are essentially the same as those of the measured loadings of Co and Mn (Table S1). The effect of Mn/Co molar ratio on the mass-specific activity and product distribution (Fig. S1) shows that the highest cobalt-time yield (CTY), selectivity of C_{2+} olefins ($C_{2+}^{=}$), $C_{5+}^{=}$, and olefin/

paraffin (O/P) ratios are achieved at a Mn/Co molar ratio of 0.2. When the reaction temperature is concerned (from 210 to 230 °C, Fig. S2a), the Co-MnO_x-(C)/SiO_2 presents significantly higher olefin selectivity together with higher CTY and CO conversion than the other three catalysts at different reaction temperatures. In addition, STY of $C_{5+}^{=}$ over Co-MnO_x-(C)/SiO_2 at 230 °C reaches as high as 0.103 $g{\cdot}g_{Cat.}^{-1}{\cdot}h^{-1}$, which exceeds that over Co/SiO_2 (0.020 $g{\cdot}g_{Cat.}^{-1}{\cdot}h^{-1}$) by a factor of 5.2 (Fig. S2b). When the GHSV increases from 2.4 to 9.6 $L{\cdot}g_{cat}^{-1}{\cdot}h^{-1}$, the following facts are clear, i.e., the lowering of CO conversion from 75% to 11%; the simultaneous increasing of $C_{2+}^{=}$ and $C_{5+}^{=}$ selectivity from 22% and 15% to 50% and 36%, respectively; the dropping of C_{5+}^{0} selectivity from 66% to 30% (Fig. S3a). If the evolution of the O/P molar ratio is examined (Fig. S3b), the molar ratio of $C_{5+}^{=}C_{5+}^{0}$ is significantly increased from 0.2 to 1.2 with increasing GHSV from 2.4 to 9.6 $L{\cdot}g_{cat}^{-1}{\cdot}h^{-1}$. These results indicate that the secondary reactions at the catalyst surface are suppressed at a higher GHSV and a lower CO conversion. Additionally, all of the counterpart catalysts show different extents of deactivation during the durability tests for a time-on-stream (TOS) of 50 h (Fig. S4a), especially for Co-(C)/SiO_2 with a fast deactivation during the initial TOS of 20 h. In contrast, Co-MnO_x-(C)/SiO_2 shows the highest CO conversion of ~80% and is stable for a TOS of 400 h (Fig. S4c). Moreover, Co-MnO_x-(C)/SiO_2 catalyst exhibits obviously higher olefin selectivity than other three counterparts even at a high CO conversion (Fig. S4b, Table S2).

To understand the intrinsic effect of MnO_x, FTS reactions were performed at 210 °C with a low iso-conversion level of 10 ± 1.4% via regulating GHSV. As revealed from the estimations (Section 1.2, Supplementary material), the diffusion limitations of reactants (syngas) are negligible under the tested conditions. Thus, the measured intrinsic activity can be safely discussed by correlating with the chemical and structural properties of the catalysts. From the reaction results (Fig. 1a and Table S3), Co-MnO_x-(C)/SiO_2 still shows the highest specific activity indexed by either CTY (2.6 $\times 10^{-5}$ $mol_{CO}{\cdot}g_{Co}^{-1}{\cdot}s^{-1}$) or turnover frequency (TOF, 0.06 s^{-1}). Moreover, clearly higher TOF and CTY over Co-MnO_x-(C)/SiO_2 than those over Co-(C)/SiO_2 indicate a positive effect of MnO_x on the catalytic activity. In contrast, the introduction of MnO_x over Co-MnO_x/SiO_2 gives negligible impact on the FTS activity as revealed from the very similar CTY ((1.5 ± 0.2) $\times 10^{-5}$ $mol_{CO}{\cdot}g_{Co}^{-1}{\cdot}s^{-1}$) or TOF (0.03 ± 0.003 s^{-1}) with that of Co/SiO_2.

In the case of the product distribution, all of the catalysts show a similar CH_4 selectivity of 8.0 ± 1.0% and ignorable CO_2 (less than 1%), indicating the unimportant impact of MnO_x. If the olefin products are taken into account, MnO_x in the case of Co-MnO_x-(C)/SiO_2 vigorously enhances the formation of olefins as indicated from the decreased selectivity of $C_{2+}^{=}$ in the order of ~56% over Co-MnO_x-(C)/SiO_2 > ~36% over Co-MnO_x/SiO_2 > > ~16% over Co/SiO_2 and ~13% over Co-(C)/SiO_2. Significantly, C_3-C_{10} olefins are dominated and $C_{5+}^{=}$ accounts for about 66% of the total olefins over Co-MnO_x-(C)/SiO_2 (Fig. 1b), leading to the much higher selectivity of $C_{5+}^{=}$ (37%) than that over Co-(C)/SiO_2 (9%). The hydrocarbon selectivity follows ASF distributions, and the calculated chain-growth probability (denoted as α_{total}, Fig. 1c) decreases in the order of Co/SiO_2 (0.87) ≈ Co-(C)/SiO_2 (0.88) > Co-MnO_x/SiO_2 (0.82) > Co-MnO_x-(C)/SiO_2 (0.76). Compared to the unprompted Co catalysts, the lower α_{total} of Co-MnO_x-(C)/SiO_2 may be originated from the significant higher proportion of olefins in the products. Eventhough the magnitude contributed from the secondary reactions of α-olefin to α_{total} of FTS products is still under debate [36,37]. A decreased α_{total} over Mn-promoted catalysts was commonly obtained under the conditions that the secondary reactions of olefins were effectively suppressed [38]. Specifically, from the calculated chain-growth probability of olefins (α_{ole}) over the four tested catalysts (Fig. S5), the $\alpha_{ole(3-8)}$ for C_3-C_8

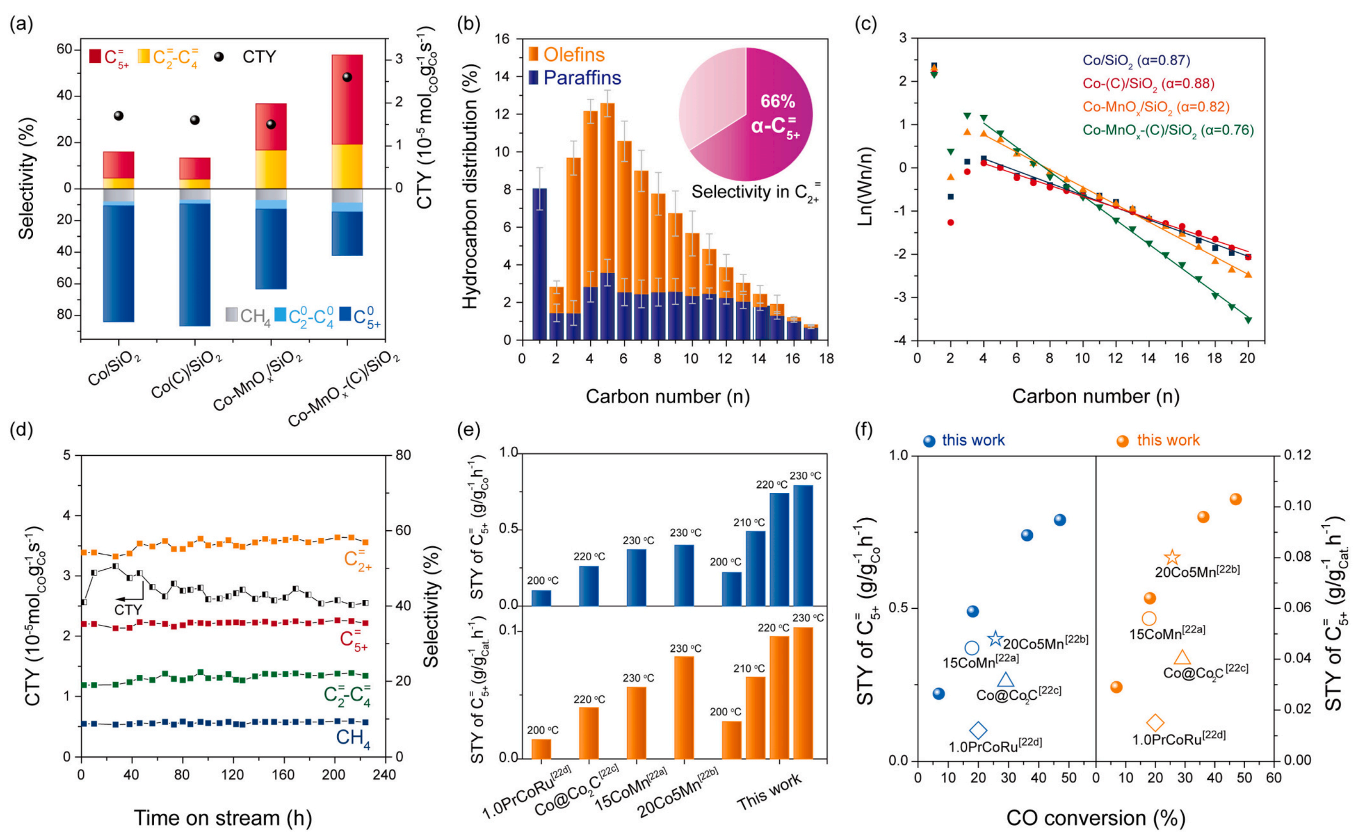


Fig. 1. Catalytic performances. (**a**) CTY and the product selectivity. (**b**) Detailed hydrocarbon distribution and selectivity of grouped olefins (inset) over Co-MnO_x-(C)/SiO_2, error bounds for the selectivity of paraffins and olefins are ± 1.1% and ± 1.2%, respectively. (**c**) ASF plots and chain growth probabilities (α_{total}) of different catalysts. (**d**) Stability results over Co-MnO_x-(C)/SiO_2. (**e**) Mass-specific productivity ($STY_{Cat.}$) and cobalt-specific productivity (STY_{Co}) of $C_{5+}^{=}$ over Co-MnO_x-(C)/SiO_2 and the published Co-based catalysts. (**f**) STY of $C_{5+}^{=}$ as a function of CO conversion over Co-MnO_x-(C)/SiO_2 and the published Co-based catalysts. Reaction conditions for (a-d): 0.50 g catalyst, 210 °C, 1.0 MPa, H_2/CO = 2.0, GHSV of 4.8 to 7.2 $L{\cdot}g_{cat}^{-1}{\cdot}h^{-1}$ for achieving the CO conversion at 10.0 ± 1.4%.

olefins are almost identical (0.75 ± 0.02). In contrast, the $\alpha_{ole(9-14)}$ for C_9-C_{14} olefins are increased in the order of Co/SiO_2 (0.57) ≈ Co-(C)/SiO_2 (0.56) < Co-MnO_x/SiO_2 (0.63) < Co-MnO_x-(C)/SiO_2 (0.70). These results clearly indicate a higher chain-growth ability of olefins on Co-MnO_x-(C)/SiO_2.

The superiority of Co-MnO_x-(C)/SiO_2 catalyst for selectively synthesizing $C_{5+}^{=}$ is further demonstrated from the FTS results with a TOS of 200 h, and the steady-state selectivity of $C_{5+}^{=}$ (37%), $C_{2+}^{=}$ (56%), and CH_4 (9%) and the CTY (2.5 $mol_{CO}\cdot g_{Co}^{-1}\cdot s^{-1}$) are kept stable without observable decrease (Fig. 1d). Thus, the long-term stability of Co-MnO_x-(C)/SiO_2 is reasonably expected. Moreover, in comparison with the published results for selective synthesis of long-chain olefins over Co-based FTS catalysts (Table S4) [39–42], Co-MnO_x-(C)/SiO_2 exhibits distinct advantages in enhancing both the catalytic activity (CTY) and the selectivity of $C_{5+}^{=}$ while lowering the undesired by-products of CH_4 and CO_2. To make a detailed comparison, the dependence of olefin selectivity on the CO conversion is given in Fig. S6 for Co-MnO_x-(C)/SiO_2, its counterpart catalysts, and those reported in references. In general, the olefin selectivity goes down with increasing CO conversion, but our Co-MnO_x-(C)/SiO_2 exhibits relatively higher olefin selectivity, especially $C_{5+}^{=}$, than that over most of the reference catalysts at the same conversion level. In addition, to quantify the productivity of $C_{5+}^{=}$, the STY of $C_{5+}^{=}$ is normalized based on the weight of the loaded catalyst (the mass-specific productivity, $STY_{Cat.}$) and the cobalt mass over the loaded catalyst (the cobalt-specific productivity, STY_{Co}), respectively. From the results given in Fig. 1e, both $STY_{Cat.}$ (0.103 $g\cdot g_{cat.}^{-1}\cdot h^{-1}$) and STY_{Co} (0.79 $g\cdot g_{Co}^{-1}\cdot h^{-1}$) of $C_{5+}^{=}$ over Co-MnO_x-(C)/SiO_2 catalyst are much higher than those reported in the references, i.e., 0.056 $g\cdot g_{cat.}^{-1}\cdot h^{-1}$ and 0.37 $g\cdot g_{Co}^{-1}\cdot h^{-1}$ for 15CoMn/SiO_2, and 0.080 $g\cdot g_{cat.}^{-1}\cdot h^{-1}$ and 0.40 $g\cdot g_{Co}^{-1}\cdot h^{-1}$ for 20Co5Mn/SiO_2 under the same reaction temperature of 230 °C. The same conclusion still holds for the FTS reaction performed at a lower temperature of 220 or 200 °C if Co/Co_2C or 1.0PrCoRu/Al_2O_3 is compared with our catalyst. Considering the following facts, i.e., varied FTS conditions from different research groups and the enhanced secondary reactions at a higher CO conversion, STY of $C_{5+}^{=}$ is plotted as a function of the CO conversion. As indicated from Fig. 1f, at the same CO conversion levels, both $STY_{Cat.}$ and STY_{Co} of $C_{5+}^{=}$ over Co-MnO_x-(C)/SiO_2 are still the highest among those over the available referenced catalysts. Therefore, the excellent catalytic performance and the essentially simple catalyst preparation method make Co-MnO_x-(C)/SiO_2 a robust and industrially important catalyst for selective synthesis $C_{5+}^{=}$ via the FTS route.

3.2. Formation of defective Co_3O_4 NPs

Apparently, glucose plays a key role for the formation of Co-MnO_x-(C)/SiO_2 with the distinguished FTS performance. Thus in situ Raman was applied to probe the carbon evolvement during carbonization and calcination stages of different precursors (i.e., Glu/SiO_2, Co-Glu/SiO_2 and Co-Mn-Glu/SiO_2 by impregnating aqueous solution of glucose, glucose and Co^{2+}, and glucose, Co^{2+} and Mn^{2+} into SiO_2, respectively) (Fig. S7). Two representative bands at around 1350 cm^{-1} (*D*-band) and 1600 cm^{-1} (*G*-band) assigning to graphitic carbon are observed over each of precursors during the carbonization process. It is noteworthy that the similar ratios of I_D/I_G obtained over Co-Glu/SiO_2 and Co-Mn-Glu/SiO_2 are higher than those over Glu/SiO_2, indicating the favorable effect of metal cations on the formation of defect-rich graphitic carbon. When these carbonaceous precursors are further calcined in air, the transformation of *G*-band to D^{**} peak occurs at above 300 °C for both Co-Glu/SiO_2 and Co-Mn-Glu/SiO_2 but not for Glu/SiO_2. This indicates that most of graphitic carbon over the precursors containing cobalt can be removed, while a tiny amount of carbon residual is still retained, which is supported by the detectable D^{**} peaks originated from the graphite fragments with low coordinated carbon atoms [43].

As indicated from results of the N_2 physisorption (Fig. S8 and Table S1), the calcined catalysts possess similar mesoporous textures. The TEM images (Fig. S9) show that all of metal oxide NPs are well dispersed on the support with a mean size of 8.0 ± 1.4 nm for Co/SiO_2, 8.2 ± 1.5 nm for Co-MnO_x/SiO_2, 6.7 ± 1.1 nm for Co-(C)/SiO_2, and 6.2 ± 1.1 nm for Co-MnO_x-(C)/SiO_2. The XRD patterns (Fig. 2a) of the calcined catalysts show the cubic Co_3O_4 phase (JCPDS 01–076-1802). Noticeably, the diffraction peaks observed over Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 are much broader than those over Co/SiO_2 and Co-MnO_x/SiO_2, indicating the lower crystallinity and smaller size of Co_3O_4 NPs on the carbon-mediated catalysts. Raman spectra show a set of representative active modes (A_{1g}+3 F_{2g}+E_g) attributing to the cubic Co_3O_4 structure [44] (Fig. 2b). The full width at half maximum (FWHM) of the A_{1g} bands on Co-(C)/SiO_2 (30 cm^{-1}) and Co-MnO_x-(C)/SiO_2 (50 cm^{-1}) are obviously broader than those of Co/SiO_2 (16 cm^{-1}) and Co-MnO_x/SiO_2 (22 cm^{-1}), suggesting an increased cation disorder and the presence of oxygen vacancies [45–47]. Moreover, it is noted that there is a red shift of A_{1g} mode from 670 cm^{-1} over Co-(C)/SiO_2 to 657 cm^{-1} over Co-MnO_x-(C)/SiO_2, which is ascribed to the increased residual stress and the lattice distortion of Co_3O_4 probably derived from the interactions between the MnO_x and the surface oxygen vacancies of Co_3O_4.

The deconvoluted XPS spectra of Co $2p_{3/2}$ (Fig. 2c) show the clearly boosted Co^{2+} peak and stronger Co^{2+} satellites over Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2, implying the generation of more surface Co^{2+} species over the carbon-mediated catalysts. The atomic ratios of Co^{2+}/Co^{3+} for Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 are 7.4 and 7.9, respectively, which are significantly higher than those of Co/SiO_2 (1.8) and Co-MnO_x/SiO_2 (2.4) (Table S5). The typical Co_3O_4 crystal consists of the tetrahedral Co^{2+} and octahedral Co^{3+} ions, and the substantial high surface Co^{2+}/Co^{3+} ratio is indicative of the presence of great amount of low-coordinated octahedral Co^{2+} ions and oxygen vacancies [48]. The lower Co chemical state of Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 is further verified by the Co K-edge X-ray absorption near-edge structure (XANES) spectra (Fig. S10). Moreover, the extended X-ray absorption fine structure (EXAFS) spectra (Fig. 2d) show two peaks with *R* space of ≈ 1.5 and ≈ 2.5 Å corresponding to the first shell Co-O and Co-Co_{long} coordinations, respectively [49]. It is obvious that Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 exhibit lower peak intensities of Co-O and Co-Co_{long} than those of Co/SiO_2 and Co-MnO_x/SiO_2, suggesting the low coordination and a low concentration of Co-O and Co-Co_{long} sites. According to the EXAFS fittings (Table S6), the Co-O coordination number (CN) of 4.0 and 4.1 for Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2, respectively, are lower than the corresponding values of Co/SiO_2 and Co-MnO_x/SiO_2 (CN of 4.6 and 4.4), indicating the existence of oxygen vacancies on Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2. The CN values of Co-Co_{long} bond for Co-(C)/SiO_2 (3.2) and Co-MnO_x-(C)/SiO_2 (4.4) are lower than those of Co/SiO_2 (5.5) and Co-MnO_x/SiO_2 (5.6). This indicates that Co particles on Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 are smaller, which are consistent with the XRD and TEM results.

3.3. Interactions between metallic Co and MnO_x

The H_2-TPR profiles (Fig. 3a) show three H_2-consumption peaks, which are corresponding to the reduction of Co_3O_4 to CoO (α), CoO to metallic Co (β), and cobalt species interacting with SiO_2 support (γ), respectively. The benchmark of supported MnO_x shows a small peak at around 300 °C ascribing to the reduction of Mn^{4+} to Mn^{2+} [50], and it is partially overlapped with the α peak of Co_3O_4 to CoO on Mn-promoted catalyst. Significantly, the γ peak is intensified and appears at a higher temperature region over Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2, which is originated from the small and highly dispersed Co NPs having stronger interactions with the support [51]. In addition, the β peak is broader and is shifted towards a higher temperature over the Mn-promoted catalysts relative to their counterparts without Mn, which is probably caused by the intimate interactions between CoO and MnO_x species [52]. The particle sizes and reducibility calculated from the results of H_2-chemisorption and O_2 titration further confirm the smaller size and lower

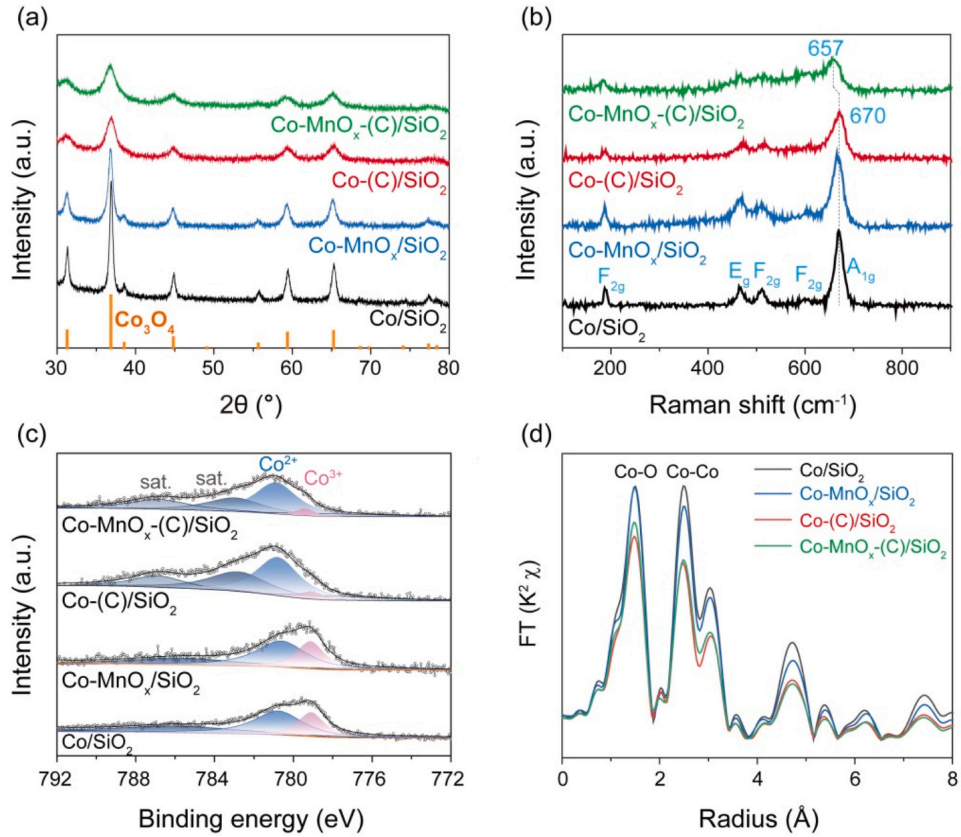


Fig. 2. Structural and chemical properties of the calcined catalysts. (**a**) XRD patterns, (**b**) Raman spectra, (**c**) XPS in Co $2p_{3/2}$ regions, and (**d**) FT-EXAFS spectra in R space for Co K-edge.

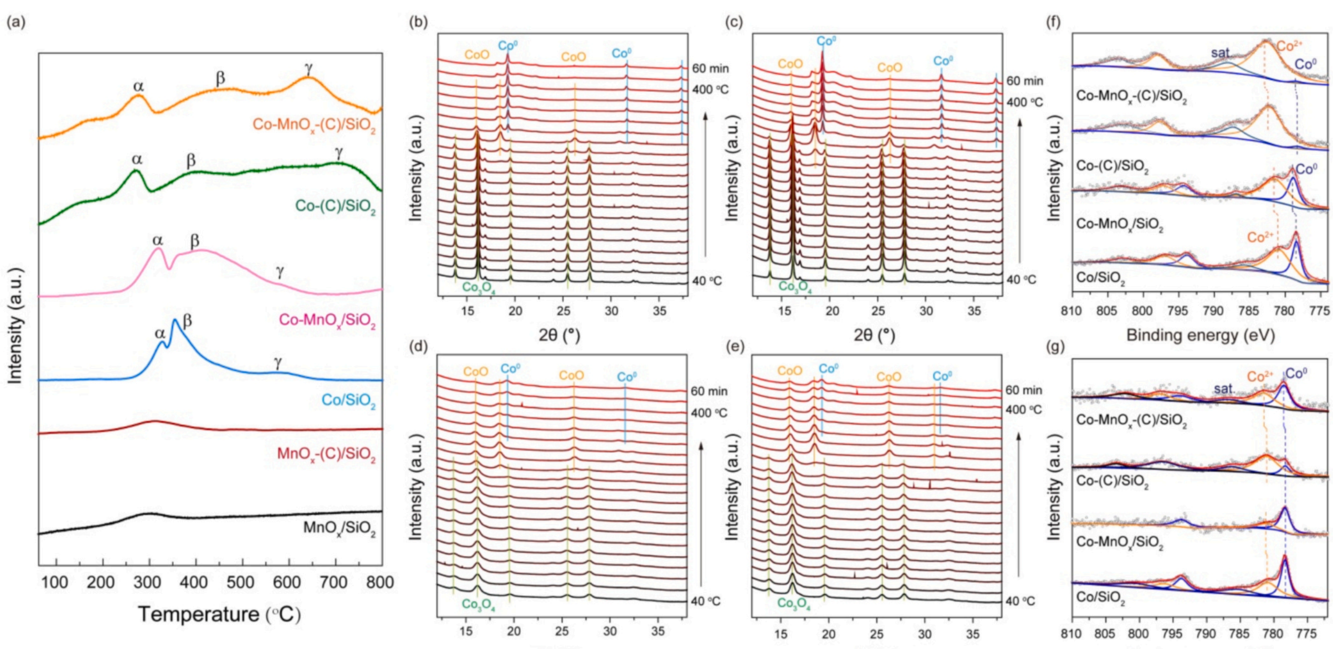


Fig. 3. Reducibility and chemical property of catalysts. (**a**) H_2-TPR profiles. *In situ* XRD patterns of (**b**) Co/SiO_2, (**c**) Co-MnO_x/SiO_2, (**d**) Co-(C)/SiO_2, and (**e**) Co-MnO_x-(C)/SiO_2 collected from 40 to 400 ºC in a flow of 10%H_2/90%Ar. Pseudo-in situ XPS spectra in Co 2p regions of the catalysts (**f**) after reduction and (**g**) after reaction.

reducibility of Co NPs on the carbon-mediated catalysts (Table S7). It is also found that the promotion of Mn decreases the catalyst reducibility but slightly influences the particle size.

In situ XRD spectra show the phase transformations from Co_3O_4 to CoO and to metallic Co (Fig. 3b-e). It is noted that the carbon-mediated catalysts show a slowness in CoO reduction, and the Mn-promotion further retards the reduction of CoO, which is more obvious on Co-MnO_x-(C)/SiO_2 with the co-existence of Co^0 and CoO under reduction at 400 °C. Given the structural properties of calcined carbon-mediated catalysts with abundant octahedral Co^{2+} ions and oxygen vacancies, it is thus proposed that the MnO_x species tend to anchor around the oxygen vacancies of octahedral Co^{2+}, which reinforces the local structure and retards the CoO reduction.

The pseudo-in situ XPS spectra of the reduced catalysts (Fig. 3f) show that the relative contents of Co^0 on Co/SiO_2 and Co-MnO_x/SiO_2 are higher than those on Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2, in consistent with the H_2-TPR and in situ XRD results. Moreover, a shift of ca. 0.4 eV to high binding energy is observed on the Mn-promoted catalysts, suggesting that the valence state of Co is more positively charged by transfering electrons to MnO_x. After FTS reaction for 10 h (Fig. 3g), the amount of Co^{2+} decreases and Co^0 increases further (Table S5). Meanwhile, it is stressed that the relative contents of Co^0 on Mn-promoted catalysts are higher than those on their counterparts, which is more pronounced in the pair of Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2. Given that the surface Co^0 species especially in small particles could be easily oxidized by water formed during FTS reaction [53,54], thus it is most likely that the interactions between Co and MnO_x in turn alleviate the oxidation of Co^0 species. In addition, it is worth noting that such a shift in binding energy is mostly diminished on two counterparts of Co-MnO_x/SiO_2 and Co/SiO_2, which is probably due to the interface segregation of Co and MnO_x during FTS reaction [32]. In contrast, the energy shift (0.4 eV) still presents in Co-MnO_x-(C)/SiO_2 as compared with Co-(C)/SiO_2, suggesting the stable interaction between Co and MnO_x.

The catalysts after reaction were further characterized to identify the active structure. XRD patterns show that metallic Co (44.2°, JCPDS 00–015-0806) is predominant on the used catalysts accompanied with weak diffractions of CoO (JCPDS 03–065-2902) (Fig. 4a). The XRD peaks over Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 are still broader relative to those over Co/SiO_2 and Co-MnO_x/SiO_2, and no Mn species are detected. However, the Mn 2p XPS spectra and Mn K-edge XANES spectra illustrate that Mn^{4+} is partially reduced to Mn^{2+} on the used

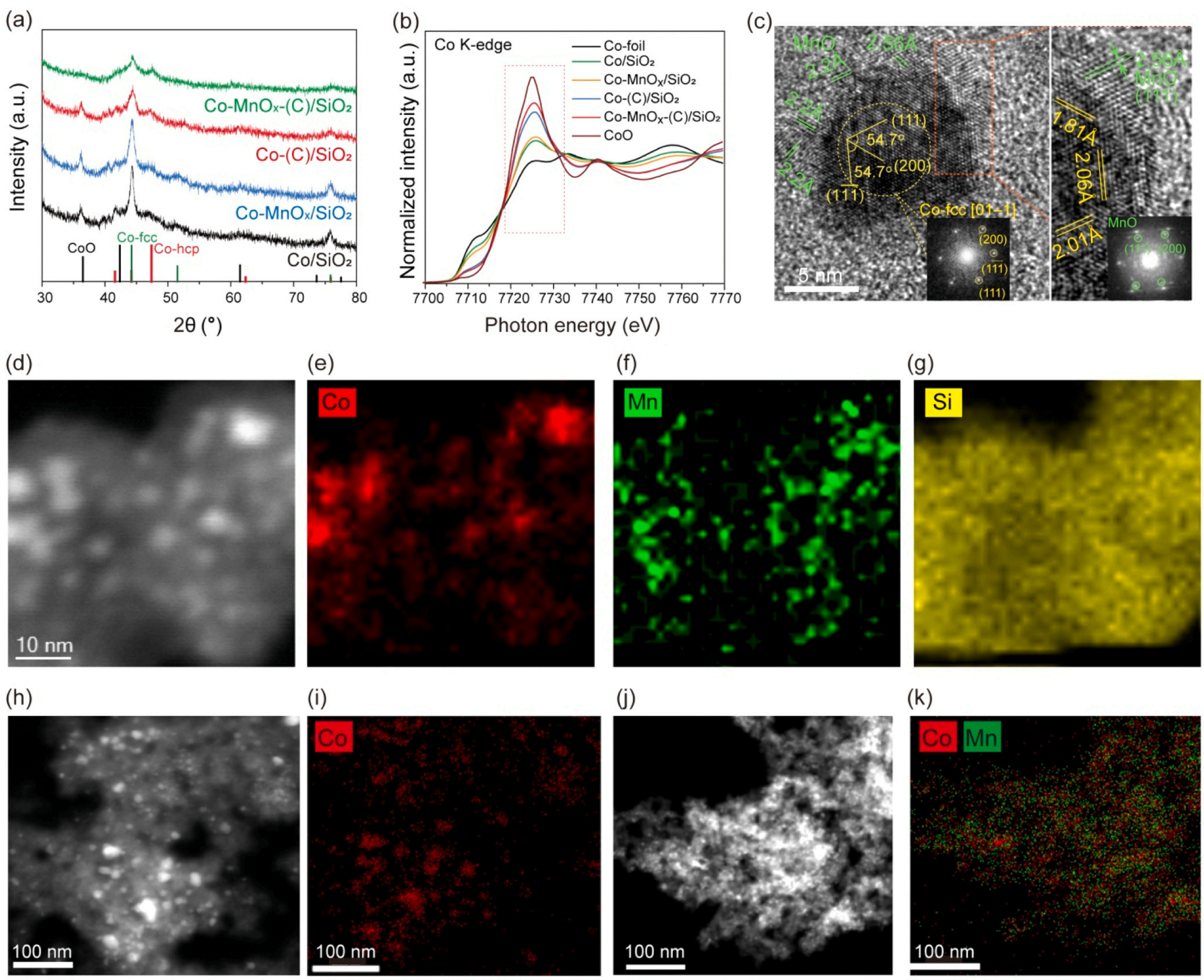


Fig. 4. Structural and chemical properties of the used catalysts. (**a**) XRD patterns. (**b**) Normalized XANES spectra at Co K-edge. (**c**) HR-TEM image of the used Co-MnO_x-(C)/SiO_2. (**d**) High-resolution HAADF-STEM images on the random thin sample specimen of the used Co-MnO_x-(C)/SiO_2 with the corresponding EDS elemental maps (**e**) for Co (red), (**f**) for Mn (green), and (**g**) for Si (yellow). HAADF-STEM images in a larger scope and the corresponding elemental mappings of the used catalysts: (**h-i**) Co-(C)/SiO_2 and (**j-k**) Co-MnO_x-(C)/SiO_2. Reaction conditions: 210 °C, 1.0 MPa, H_2/CO= 2.0, TOS of 50 h.

catalysts (Fig. S11). According to the Co K-edge XANES spectra (Fig. 4b), the absorption edges of Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 are more close to the standard CoO, suggesting the high content of CoO. Moreover, the intensity of white line on Co-MnO_x-(C)/SiO_2 is significantly higher than that on Co-(C)/SiO_2, suggesting the pronounced electron transfer from Co to Mn. However, this is not obvious on the two counterparts of Co-MnO_x/SiO_2 and Co/SiO_2.

According to the TEM images of the used catalysts (Fig. S9), Co/SiO_2 and Co-(C)/SiO_2 exhibit obvious sintering with enlarged particle-size distributions. This phenomenon, however, is not apparent in the cases of the Mn-promoted catalysts, especially for Co-MnO_x-(C)/SiO_2, which still retains well dispersed Co NPs having a narrow size distribution similar to those of the calcined one. Moreover, the HR-STEM images of the used Co-MnO_x-(C)/SiO_2 clearly show the close spatial proximity of Co and Mn species (Fig. 4c and Fig. S12), i.e., MnO_x in the form of nanoclusters are well dispersed and attached around the metallic Co NPs, as evidenced by the lattice fringes of 2.20 Å and 2.56 Å assigning to (111) and (200) of MnO, respectively. No obvious spatial segregation occurs on the used Co-MnO_x-(C)/SiO_2 as compared to that on Co-MnO_x/SiO_2 (Fig. S12). The spatial distributions of Mn species and Co nanoparticles were further characterized by the high-magnification HAADF-STEM and corresponding X-ray energy-dispersive spectroscopy (EDS) mapping on the random thin sample specimens of the used Co-MnO_x-(C)/SiO_2 (Fig. 4d-g and Fig. S13). Clearly, Mn is intimately adjacent to Co NPs, while the phenomena of Mn-Co segregation and Mn-migration to the support surface are not observed. The HAADF-STEM image and the corresponding EDS mapping in a larger scope confirm the highly dispersed particles with homogeneous Co and Mn elements over the used Co-MnO_x-(C)/SiO_2 relative to the big particles and aggregated Co elements over the used Co-(C)/SiO_2 (Fig. 4h-k). It is thus reasonable to speculate that the stable Co-MnO_x interaction can effectively resist the sintering of Co NPs.

3.4. Insights into the formation of long-chain olefins

In situ DRIFTS spectra of CO adsorption on the reduced catalysts were collected in a flow of 30%CO/Ar with increasing the temperature (Fig. S14). At 40 ºC, two bands at 2032 and 2013 cm^{-1} ascribed to the CO linear adsorption on the top site of metallic Co (CO_{top}) are developed on the four tested catalysts, primary differences of which are higher intensities over the carbon-mediated catalysts. When the temperature is increased, the CO_{top} bands are diminished, which is most likely due to the CO desorption and/or dissociation. Considering that the carbon-mediated catalysts with small Co NPs have a great amount of step-edge and defect sites, the CO direct dissociation would occur below the CO desorption temperature [55,56]. Furthermore, at the temperature of 220 ºC, a new CO_{top} peak evolves at ~2055 cm^{-1} for Co/SiO_2 and Co-MnO_x/SiO_2. This identical CO_{top} peak (~2060 cm^{-1}) arise on Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 at 170 ºC, and it is intensified and moves gradually to higher wavenumbers until 2072 cm^{-1} when the temperature is increased up to 220 ºC. The blue-shift of the CO_{top} peak with an increased intensity is from the improved CO coverage and the presence of co-adsorbed O or C [57,58], leading to a decreased electron back-donation from the metal site to the 2π* orbital of CO. It is thus expected that the catalysts of Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 have an enhanced capability for the CO adsorption and direct dissociation. Additionally, if Co-(C)/SiO_2 and Co-MnO_x-(C)/SiO_2 catalysts with comparable particle size distributions (Table S7) are compared, the CO_{top} locations and their deriving trends with increasing the temperature are basically similar during CO adsorption experiments (Fig. S14g-h). The most noticeable change of CO_{top} bands upon Mn addition is the reduced intensity, indicating a loss of active sites for the CO adsorption due to masking effect of MnO_x, which is in accordance with the characterization of H_2 chemisorption [59]. Likewise, the set of Co/SiO_2 and Co-MnO_x/SiO_2 counterparts show the same evolution phenomenon (Fig. S14e-f). Another distinguishing feature of Mn-promoted catalysts is the adsorption band at 1589 cm^{-1} attributed to the asymmetric *O-C-O stretching vibration [60,61], which is probably caused by the CO dissociation and disproportionation on Co-MnO_x interfaces, keeping inconsistent with the previous conclusion proposed by Johnson et al. [62].

Fig. 5a-d show in situ DRIFTS spectra of CO hydrogenation on the reduced catalysts with increasing temperature. The bands around 2030, 1940, and 1780 cm^{-1} are attributed to CO adsorption on top site (CO_{top}), CO bridge-bounded (CO_{bridge}, ~1940 cm^{-1}) and CO located on threefold hollow site (CO_{hollow}, ~1780 cm^{-1}) [58], respectively. Firstly, the CO_{top} on the four catalysts is varied in intensity and frequency with increasing the temperature, and the CO_{top} frequency as a function of the temperature is shown in Fig. 5e. The CO_{top} of the Mn-promoted catalysts locate at the higher wavenumbers than those of their corresponding Co-based catalysts at temperature below 180 ºC. When the temperature is increased (>180 ºC), CO_{top} of Co-MnO_x/SiO_2 shifts rapidly to the lower wavenumbers similar to that of Co/SiO_2, accompanied with a quick decrease in the IR intensity. In contrast, the CO_{top} of Co-MnO_x-(C)/SiO_2 always stays about 10 cm^{-1} wavenumbers higher than that of Co-(C)/SiO_2 throughout the whole test. Upon the introduction of syngas, the competitive adsorption between CO and H_2 occurs on the catalyst surface, leading to a decrease in the CO adsorption. Concurrently, the co-absorbed H contributes to the electron density of the Co metal, which increases the extent of the π * back-donation to CO, resulting in a red shift of the CO_{top} peak. Given the identical locations of CO_{top} bands with 30%CO/Ar on each pair of catalysts (Fig. S14e-h), the CO_{top} peak on Co-(C)/SiO_2 displays a greater redshift than that on Co-MnO_x-(C)/SiO_2 in the presence of H_2 (Fig. 5e), suggesting a significant inhibiting effect of MnO_x on the co-adsorption of H_2 over Co-MnO_x-(C)/SiO_2 [63]. Moreover, this effect proves to be more enduring on Co-MnO_x-(C)/SiO_2 than that on Co-MnO_x/SiO_2 throughout the entire hydrogenation process. These findings support the mechanism that Mn interacting with Co particles faciliates the CO adsorption and dissociation while inhibiting the H_2 competitive adsorption, leading to an increased CO coverage over the catalyst surface. Therefore, the plentiful and stable Co-MnO_x interfaces on the Co-MnO_x-(C)/SiO_2 account for the consistent higher surface CO* (C*)/H* ratio than that on Co-(C)/SiO_2 under the FTS reaction conditions. Furthermore, Co-MnO_x-(C)/SiO_2 exhibits a substantially stronger *O-C-O peak at 1589 cm^{-1} than Co-MnO_x/SiO_2, highlighting the stability and abundance of Co-MnO_x interfaces.

The C-H stretching regions at 3015, 2930, and 2858 cm^{-1} corresponding to the asymmetric and symmetric stretching modes of methylene (-CH_2) group [64] are developed when temperature reaches 180 ºC, suggesting the occurrence of hydrogenation and the formation of hydrocarbon products. The intensities of the major adsorbate patterns plotted as a function of temperature (Fig. 5f-i) exhibit the evolvements of the surface adsorption species and intermediates during the hydrogenation process. The intensities of the CO_{top} and *O-C-O exhibit the obvious volcano trends with increasing temperature, while the intensity of the CO_{hollow} signal hardly changes or monotonously increases with the formation of *CH_x species, suggesting that the CO_{top} and *O-C-O species are mainly responsible for the formation of hydrocarbons during the CO hydrogenation. Moreover, the significantly boosted signals of both adsorbed CO* /*O-C-O and *CH_x over Co-MnO_x-(C)/SiO_2 are indicative of the improved CO adsorption and the enhanced CO hydrogenation rate, which is in good consistent with the reaction results (Fig. 1a). Further, the synchronous products are monitored by the online MS (Fig. S15). Compared to the sole major CH_4 signal over Co/SiO_2 and Co-(C)/SiO_2, the propylene signal emerged on Mn-promoted catalysts and is greatly boosted on Co-MnO_x-(C)/SiO_2, indicating excellent ability for producing olefins.

3.5. Understanding the catalytic nature

The carbon-mediated strategy facilitates the formation of highly

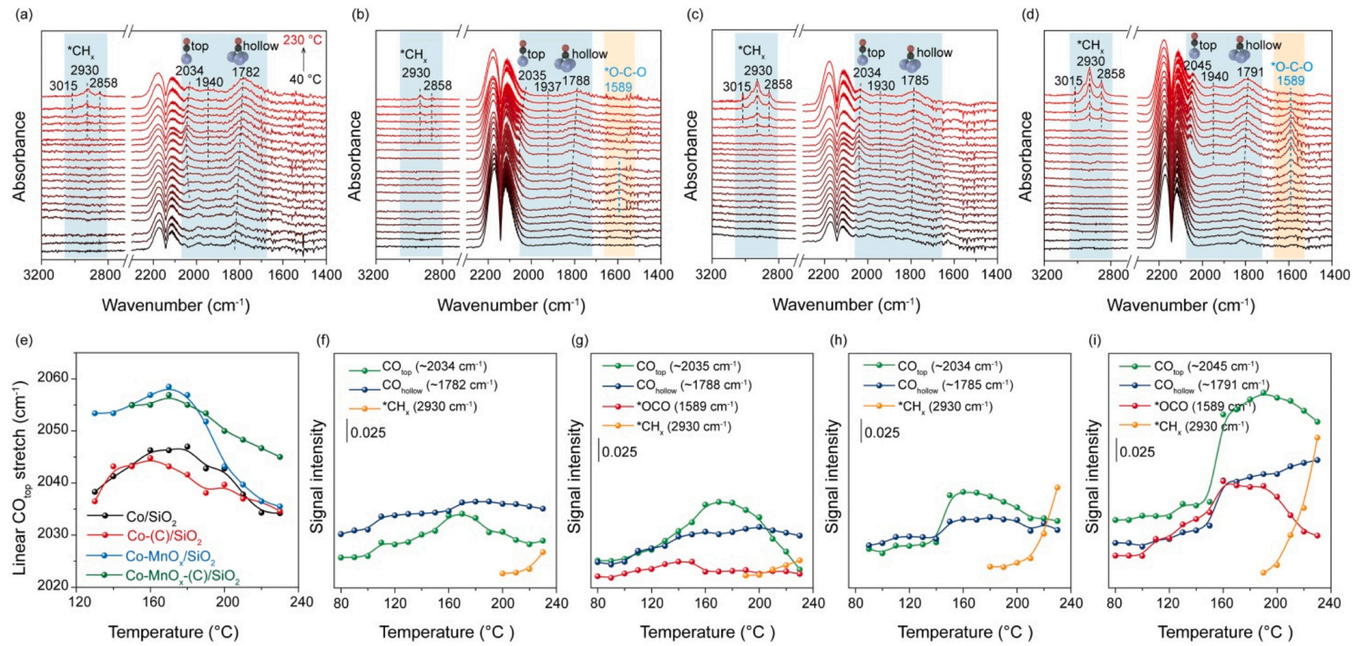


Fig. 5. Adsorbed species and reaction intermediates of FTS over the reduced catalysts. *In situ* DRIFTS collected over (**a**) Co/SiO_2, (**b**) Co-MnO_x/SiO_2, (**c**) Co-(C)/SiO_2, and (**d**) Co-MnO_x-(C)/SiO_2 from 40 to 230 °C in a flow of syngas. (**e**) The evolutions of the linear CO_{top} stretching frequency on temperature over the four catalysts. The changing profiles of the peak intensity for the main adsorbed species and intermediates over (**f**) Co/SiO_2, (**g**) Co-MnO_x/SiO_2, (**h**) Co-(C)/SiO_2, and (**i**) Co-MnO_x-(C)/SiO_2.

dispersed Co particles with an optimal size of ~6 nm. Moreover, the carbon-mediated procedure creates enriched octahedral Co^{2+} sites and oxygen vacancies on Co_3O_4 nanoparticles, which favor the deposition of MnO_x islands with stable and maximized Co-MnO_x perimeter. The resulting Co-MnO_x-(C)/SiO_2 catalyst after reaction still shows strong electronic interactions between Co and Mn as evidenced by the XPS peak shift (Fig. 3g). The stable Co-MnO_x nano-interfacial structure is identified by the HR-STEM and HAADF-STEM images (Fig. 4 and Fig. S12-13). The Co-MnO_x-(C)/SiO_2 catalyst exhibits an excellent activity and stability for the FTS reaction (Fig. 1). Although the surface Co^0 sites are partially covered by MnO_x species, both CTY and TOF of Co-MnO_x-(C)/SiO_2 are still significantly higher than those of the other catalysts (Table S3). As evidenced by the in situ DRIFTS results of CO hydrogenation (Fig. 5), i.e., Co-MnO_x-(C)/SiO_2 displays significant high intensity of absorbed CO* and *CH_x bands. These suggest the importance of optimized Co^0-MnO_x nano-interfaces for enhancing the reaction activity, which is in consistent with the mechanism that the Mn^{2+} cations exposed at the edge of Co^0-MnO_x interface may function as Lewis-acids for enhancing CO adsorption and dissociation [62]. Additionally, the dissociated O* interacting with Mn^{2+} can be removed rapidly via the formation of H_2O, resulting in the fast release of the active sites. The quick removal of in situ generated H_2O can also decrease the possibility for the potential oxidation of metallic Co, which is supported by the pseudo-in situ XPS spectra (Fig. 3g). Thus, the catalyst exhibits excellent activity and stability even at a high CO conversion of ~80% for more than 400 h (Fig. S4).

Considering that the stable and plentiful Co-MnO_x nano-interfaces ove Co-MnO_x-(C)/SiO_2 can effectively decrease the availability of surface active sites involved in hydrogen adsorption and activation (H_2-chemisorption, Table S7), while faciliate CO chemsorption and dissociation, the catalyst thus possesses substantial high surface coverage ratio of CO* (C*)/H* under the FTS reaction conditions as suggested by the in situ DRIFTS results (Fig. 5e). Moreover, the quick hydrogenation of O* species interacting with Mn^{2+} around Co-MnO_x interface tend to drive H* migration and further decrease the H* concentration surrounding the formed hydrocarbons on Co sites. Since the surface coverages of intermediates greatly influence the elementary steps [65], the high surface coverage ratio of CO* (C*)/H* is expected to enhance the rate of carbon-chain growth and β-H elimination while suppressing olefin hydrogenation, resulting in good selectivity of long-chain olefins as evidenced by the high signal of propylene form synchronous MS data (Fig. S15d).

In a proof-of-concept experiment (Fig. S16), the H_2/CO ratio in the gas feed was decreased from 2 to 0.5. Results show that the low H_2/CO ratio leads a two-fold increase in selectivity of olefins on both Co/SiO_2 and Co-(C)/SiO_2, suggesting an effective influence of tuning H_2/CO ratio on olefin production. In contrast, such increase in selectivity of olefins is not much pronounced on the Mn-promoted catalysts, especially for Co-MnO_x-(C)/SiO_2. This is because the Co-MnO_x nano-interfaces have already been endowed with the high surface coverage ratio of CO* (C*)/H* irrespective of the feed gas H_2/CO ratio. Moreover, the pulse experiments of propene (Fig. S17) show that the hydrogenation of olefins over Co-MnO_x-(C)/SiO_2 is less active than that over Co-(C)/SiO_2. This result agrees with the density functional theory (DFT) [66,67] that the manipulation of surface electronic property of Co by decorating with MnO_x is prone to inhibit the hydrogenation of olefins and promote olefin desorption.

4. Conclusion

In summary, the simple carbon-mediated strategy is developed to intensify the interaction between MnO_x and Co NPs via formation of defective Co_3O_4 with abundant O-deficient octahedral Co^{2+} cations and oxygen vacancies. The characterizations suggest that MnO_x species tend to anchor around the oxygen vacancies of these octahedral Co^{2+}, resulting in the plantiful and stable Co-MnO_x nano-interfaces during the reduction and under FTS reaction conditions. The in situ spectroscopic data further demonstrates that the Co-MnO_x-(C)/SiO_2 catalyst with the stable Co-MnO_x interfaces exhibits a high surface coverage ratio of CO* (C*)/H* and thus faciliates olefins production. The resulting catalyst presents as high as of 37% selectivity of $C_{5+}^{=}$ and 19% selectivity towards $C_2^{=}$–$C_4^{=}$ at 210 ºC with an improved catalytic activity, and cobalt based STY of $C_{5+}^{=}$ as high as of 0.79 $g\cdot g_{Co}^{-1}\cdot h^{-1}$ is avhied at 230 °C. Moreover, the optimal catalyst shows an excellent stability for over 400 h as a result of the stable Co-MnO_x nano-interfaces with the suppressed sintering of Co NPs. Thus, the straightforward strategy of the carbon-mediated surface-engineering method may open up new opportunities in fabricating and tuning the nano-interfacial structure of the supported catalysts.

CRediT authorship contribution statement

Liu Zhong-Wen: Writing – review & editing, Supervision, Resources, Funding acquisition. **Zhu Kaiyu:** Data curation. **Shui Mei-Ling:** Data curation. **Du Ning:** Data curation. **Yan Yi-Fan:** Data curation. **Song Yong-Hong:** Data curation. **Bao Jun:** Resources, Data curation. **Yang Guo-Qing:** Data curation. **Niu Yiming:** Resources, Data curation. **Liu Xingwu:** Writing – original draft, Formal analysis, Data curation. **Ma Ding:** Writing – review & editing, Resources. **Zhu Min-Li:** Writing – original draft, Investigation, Formal analysis, Data curation. **Liu Zhao-Tie:** Data curation. **Zhang Bingsen:** Resources, Data curation.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhong-Wen Liu has patent #ZL2018113369370 licensed to Shaanxi Normal University. Zhong-Wen Liu has patent #ZL2018115677639 licensed to Shaanxi Normal University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.apcatb.2024.123783.

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Supplementary material

Long-chain α-olefins production over Co-MnO_x catalyst with optimized interface

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Table of Contents

1. Experimental methods···**S3**
1.1 Evaluation of catalytic performance in FTS···S3
1.2 Mass transport limitation···S14
2. Supplementary Figures···**S17**
Fig. S1···S17
Fig. S2···S18
Fig. S3···S19
Fig. S4···S20
Fig. S5···S21
Fig. S6···S22
Fig. S7···S23
Fig. S8···S24
Fig. S9···S25
Fig. S10···S26
Fig. S11···S27
Fig. S12···S28
Fig. S13···S29
Fig. S14···S30
Fig. S15···S31
Fig. S16···S32
Fig. S17···S33
3. Supplementary Tables···**S34**
Table S1···S34
Table S2···S35
Table S3···S36
Table S4···S37
Table S5···S38
Table S6···S39
Table S7···S40
4. Supplementary References···**S41**

1. Experimental methods

1.1 Experimental details of the FTS tests and product analyses

The schematic diagram of the reaction system for the FTS experiments along with the online analysis devices is given in Fig. SE1. The FTS reactions were performed in a high-pressure stainless-steel fixed-bed reactor with an inner diameter of 9 mm. Typically, 0.5 g catalyst (40-60 mesh) diluted with an equal volume of quartz sands (40-60 mesh) was loaded at the isothermal zone of the reactor. Before the FTS test, the catalyst was reduced at 400 °C and 0.1 MPa for 5 h in a pure H_2 flow of 50 mL·min^{-1}. After the reduction, the reactor was cooled to 120 °C in the same H_2 flow. Then, the syngas with the volumetric composition of 32% CO, 64% H_2, and 4% Ar as the internal standard was switched, and pressurized to 1.0 MPa. The catalytic test was conducted at the reactor temperature range from 210 to 230 °C and gas hourly space velocity (GHSV) from 2.4 to 9.6 L·g$_{cat}^{-1}$·h^{-1}. The tail gas of the reactor was depressurized before entering an online gas chromatograph (GC), and the entire pipeline was insulated at 300 °C to avoid the product condensation. The heated flow was firstly introduced to a HP-PONA capillary column via a six-way valve, and the hydrocarbons (C_1–C_{18}) were seperated and analyzed using an FID. The representative chromatograms are given in Fig. SE2 and SE3. The pipelines, valves, and chromatographic injection port were insulated at 300 °C, and the temperature of the HP-PONA capillary column was increased from 25 to 280 °C at a ramp of 10 °C/min and held at 280 °C for 20 minutes. The flow out of the six-way valve was cooled in a cold trap (0 °C) to collect liquid products. The CO, CH_4, Ar, and CO_2 gases in the effluent from the cold trap were separated using the PROPAK Q and CST-TDX-01 packed columns at 50 °C and analyzed on a thermal conductivity detector (TCD), and the typical chromatograms are shown in Fig. SE4. All the catalytic results were obtained at a time on stream of no less than 24 h and each catalyst was tested at least for three times for calculating the experimental errors. After the reaction, the reactor was purged with N_2 (200 mL/min) for 15 min and cooled naturally in the N_2 flow to the room temperature (20-25 °C). Then the catalyst was passivated for 1 h in a 1 vol % O_2/Ar flow of 20 mL/min before unloading from the reactor. The unloaded spent catalyst was sealed in the polyester sample tube after eliminating the air in a glove box under an Ar protection.

The catalytic activity is expressed as the CO conversion, cobalt-time yield (CTY) and turnover frequency (TOF), respectively. The CO conversion (X_{CO}) is defined according to Eq. S1, where $F_{CO,in}$ and $F_{CO,out}$ are the molar flow rate of CO at the inlet and outlet of the reactor, respectively.

CTY is defined as moles of CO converted to hydrocarbons over per gram of cobalt per second as expressed in Eq. S2, where m_{Co} is the mass of cobalt in the reactor.

TOF is defined according to Eq. S3, where the "*moles* of exposed Co^0 atoms" is calculated based on the H_2-chemisorption.

The selectivity of hydrocarbon (S_{CxHy}) is defined according to Eq. S4, where F_{CxHy} represents the molar flow rate of the hydrocarbon at the outlet of the reactor.

To reveal the accuracy of the online GC analyses and determine the carbon balance of the FTS experiment, longer test (TOS>100 h) was performed over Co-MnO_x-(C)/SiO_2 catalyst under the conditions of 210 °C and 7.2 $L \cdot g_{cat}^{-1} \cdot h^{-1}$ with CO conversion of ~10%. The products were collected in the cold trap under the steady state (after a TOS of ~24 h). After statically leaving the sample tube for about 1 h, the products were spontaneously separated into an aqueous phase and an oil phase. To determine the amount of oxygenate products, the aqueous phase sample was dissolved in acetone (HPLC grade, Sigma Aldrich) and analyzed on an offline GC (Shimadzu 2010) equipped with a capillary column (DB-5, 30 m × 0.32 mm) and an FID. From the representative chromatogram (Fig. SE5), C_{1-4} alcohols were detected and their molar concentrations in the aqueous phase were determined by the external standard method. The selectivity of oxygenate ($S_{oxy.}$) was calculated according to Eq. S5, where $n_{oxy.}$ represents the moles of carbon in the oxygenates within the aqueous phase, and $n_{CO,consumed}$ represents the moles of consumed CO during the product collecting process. The selectivity of C_{1-4} alcohols was determined to be less than 2%. This is consistent with the common fact that oxygenates are much less produced for the Co-based catalysts under FTS conditions with a low pressure.

$$S_{oxy.}=\frac{n_{oxy.}}{n_{CO,consumed}}\times 100\,\% \quad (Eq.S5)$$

After a strong blending, a small amount of the oil sample was dissolved in 15.0 mL CS_2 (A.R. grade, Aladdin). The hydrocarbons dissolved in CS_2 were subjected to a full analysis on an offline GC (Agilent 7890B) equipped with a capillary column (RIPP GSD-1, 10 m × 530 μm × 0.15 μm), FID, and an analysis software of the simulated distillation method (Provided by the Sinopec Research Institute of Petroleum Processing Co., Ltd.). The temperature of the column furnace was programmed to rise from 50 to 430 °C at a temperature ramp of 10 °C/min and hold at 430 °C for 20 minutes. The representative FID signals of the oil phase are displayed in Fig. SE6. Since that the selectivity of C_7 hydrocarbons calculated based on the online GC analyses is reasonably the same to that based on the offline GC analyses of the oil products, the selectivity of hydrocarbons in the collected oil is determined. Results indicate that the discrepancy for the selectivity of C_5-C_{18} hydrocarbons between the online and offline GC analyses is below ± 3%. To confirm the accuracy, the FTS reaction was also operated for a higher CO conversion of ~30% under the conditions of 220

°C and 4.8 L·g_{cat}^{-1}·h^{-1}. The representative FID signals of the oil phase are displayed in Fig. SE7. The selectivity of $C_{1\text{-}4}$ alcohols is still below 2% and the difference in the selectivity of C_5-C_{18} hydrocarbons determined based on the online and offline GC analyses is still within ± 3%. Thus, the online GC analyses of the hydrocarbons with a carbon number up to 18 are practically accurate within the experimental errors.

Based on the FTS results with the full analyses of the condensed products, the carbon balances were also estimated according to Eq. S6. The carbon balance is defined as the total amount of carbon in all of the products divided by the amount of converted CO. Since that the selectivity of FTS products is calculated based on carbon number, the carbon balance can also be expressed as the sum of the selectivity of CO_2, C_1-C_5 hydrocarbons based on the online GC analyses, and C_{6+} hydrocarbons and the oxygenates based on offline GC analyses. From the FTS results obtained under the two sets of conditions, i.e., 210 °C and 7.2 L·g_{cat}^{-1}·h^{-1} vs. 220 °C and 4.8 L·g_{cat}^{-1}·h^{-1}, the carbon balances are reproducibly determined to be within 100 ± 10%. Thus, the experimental errors of the FTS reaction used in our case are accurate within 100 ± 10% of the carbon balances.

$$C_{balance} = \frac{n_{CO_2} + n_{HC} + n_{oxy.}}{n_{CO,consumed}} \times 100\%$$

$$\text{¿}\, S_{CO_2,online} + S_{C1-C5,online} + S_{C6+,offline} + S_{oxy.,offline} \left(Eq.S6\right)$$

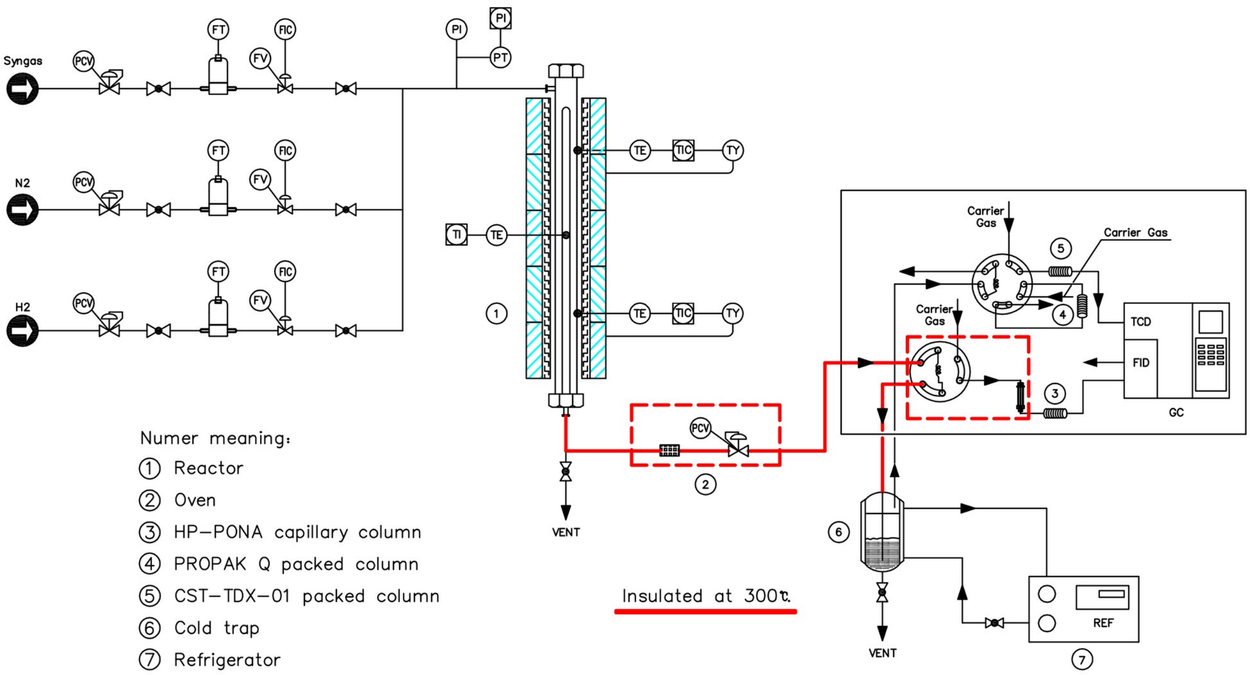

Fig. SE1. Schematic diagram of the FTS reaction system including the online GC analysis.

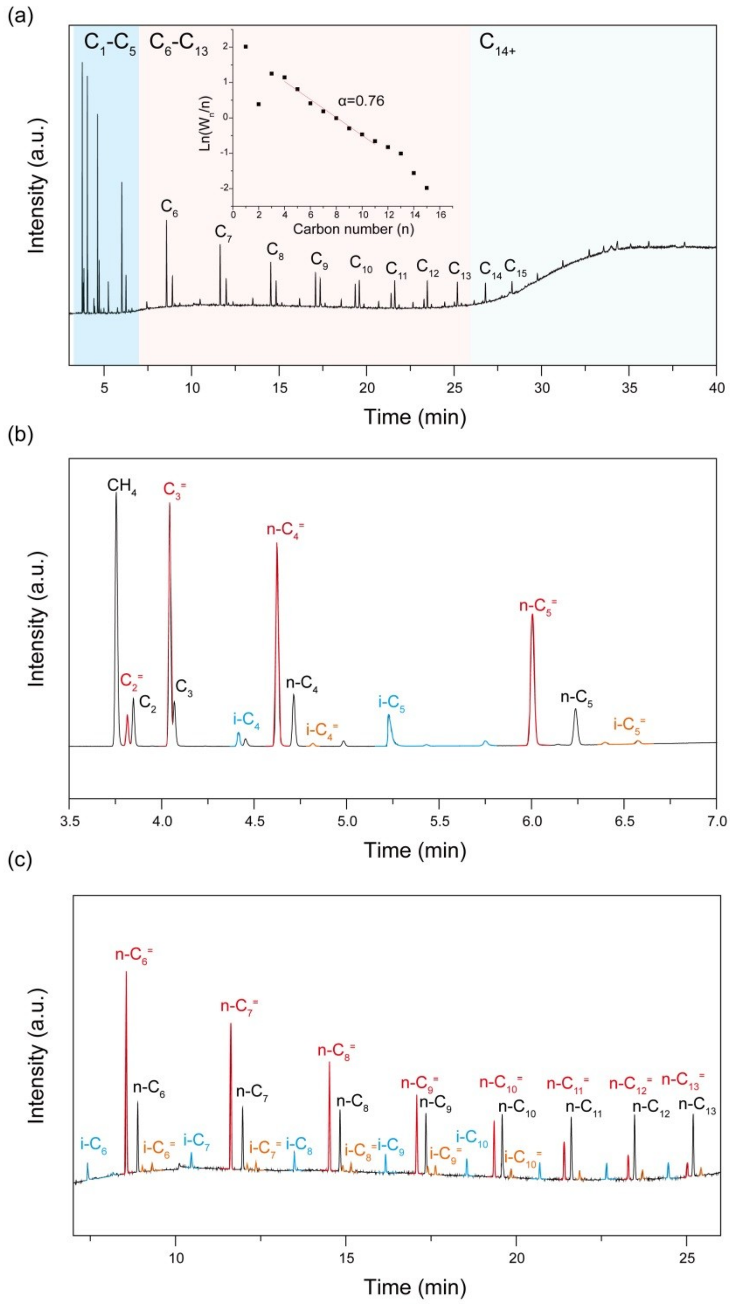

Fig. SE2. Chromatogram of hydrocarbon products from a HP-PONA capillary column. (**a**) FID signals of a representative online hydrocarbon products from a fixed-bed FTS catalytic test of Co-MnO_x-(C)/SiO_2 at CO conversion of ~10%. Inset: ASF plot and the chain growth probability (α = 0.76) obtained in the C_3-C_{16} hydrocarbon range. Detail view of hydrocarbons: (**b**) C_1-C_5, and (**c**) C_6-C_{13}. Color code: black (*n*-paraffin products, *n*-C_n), red (*n*-olefin products, *n*-$C_n^=$), blue (isomeric paraffin products, *i*-C_n), orange (isomeric olefin products, *i*-$C_n^=$). Reaction conditions: 210 °C, 1.0 MPa, GHSV=7.2 L·g_{cat}^{-1}·h^{-1}, H_2/CO = 2 and the data is collected after more than 24 h on stream.

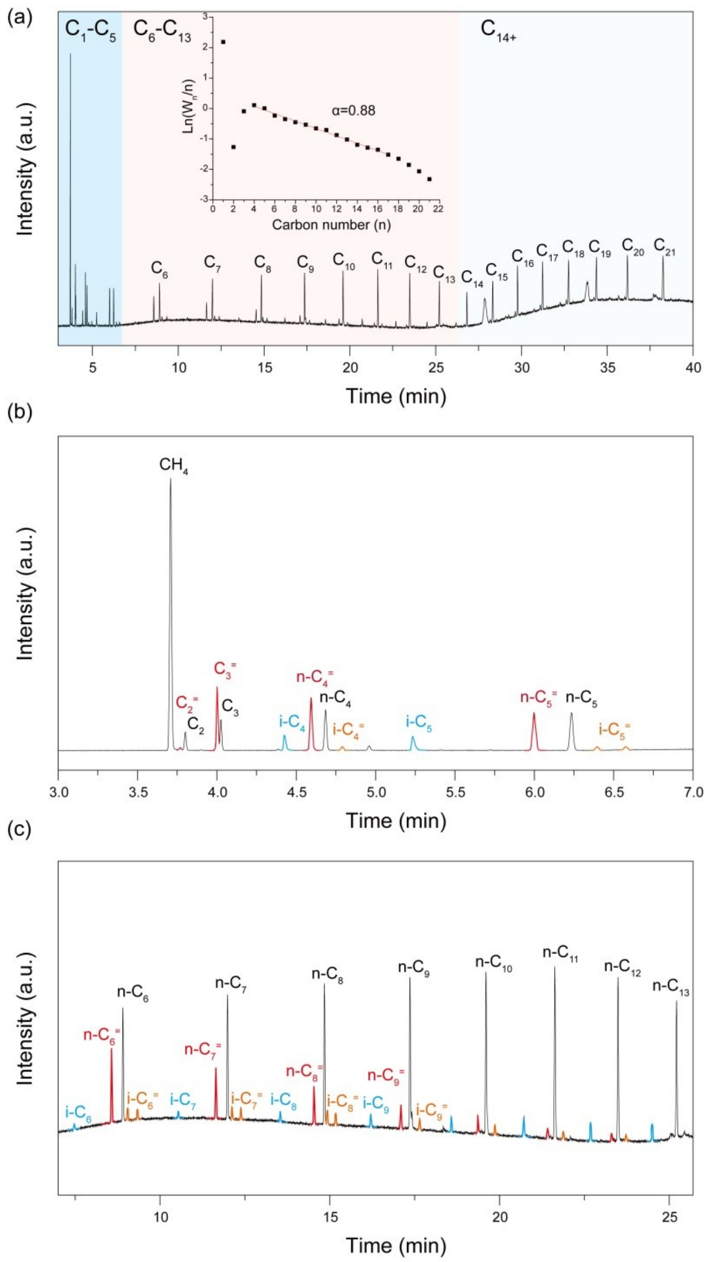

Fig. SE3. Chromatogram of hydrocarbon products from a HP-PONA capillary column. (**a**) FID signals of a representative online hydrocarbon products from a fixed-bed FTS catalytic test of Co-(C)/SiO_2 at CO conversion of ~10%. Inset: ASF plot and the chain growth probability (α = 0.88) obtained in the C_3-C_{16} hydrocarbon range. Detail views of hydrocarbons: (**b**) C_1-C_5, and (**c**) C_6-C_{13}. Color code: black (*n*-paraffin products, *n*-C_n), red (*n*-olefin products, *n*-$C_n^=$), blue (isomeric paraffin products, *i*-C_n), orange (isomeric olefin products, *i*-$C_n^=$). Reaction conditions: 210 °C, 1.0 MPa, GHSV=4.8 $L\cdot g_{cat}^{-1}\cdot h^{-1}$, H_2/CO = 2 and the data is collected after more than 24 h on stream.

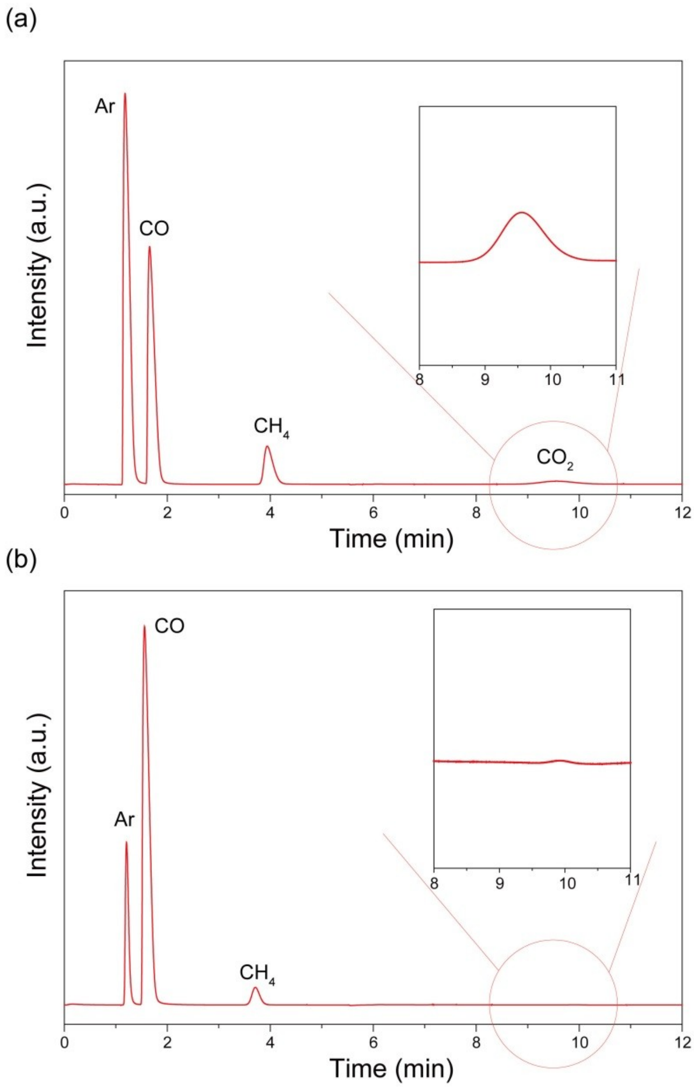

Fig. SE4. TCD signals of permanent gases eluting from the PROPAK Q and CST-TDX-01 packed columns. (**a**) TCD signals of the standard mixed gases of Ar (30%), CO (20%), CH_4 (5%), and CO_2 (2%). (**b**) TCD signals of a representative online test of Co-MnO_x-(C)/SiO_2 at CO conversion of ~40%. The invisible CO_2 signal suggests the negligible CO_2 selectivity (<1%). Reaction conditions: 220 °C, 1.0 MPa, GHSV = 4.8 $L\cdot g_{cat}^{-1}\cdot h^{-1}$, H_2/CO =2.

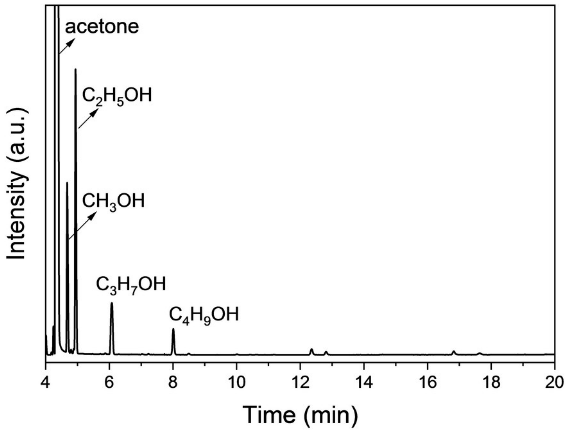

Fig. SE5. The representative chromatogram of the aqueous phase products obtained from the offline GC analysis.

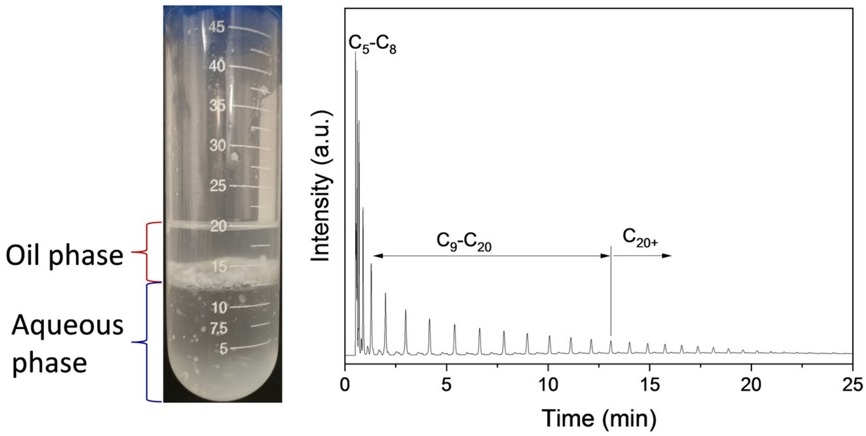

Fig. SE6. FTS products collected from a stability test over Co-MnO_x-(C)/SiO_2 at 210 °C, 7.2 L·g_{cat}^{-1}·h^{-1} at the quasi steady-state conditions (after 24 h on-stream) (left). The FID signals of the oil phase products obtained from the offline GC analysis (right).

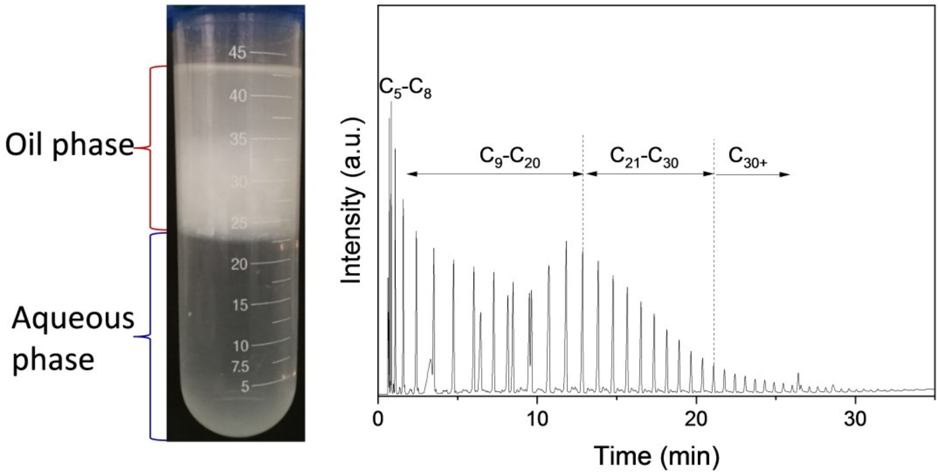

Fig. SE7. FTS products collected from a stability test over Co-MnO_x-(C)/SiO_2 at 220 °C, 4.8 L·g_{cat}^{-1}·h^{-1} at the quasi steady-state conditions (after 24 h on-stream) (left). The FID signals of the oil phase products obtained from the offline GC analysis (right).

1.2 Mass transport limitation

The influence of the interphase diffusion limitation of reactants on the tested catalysts with particle diameter of 350 μm was estimated based on the reaction results in Table S2 (220 °C) and Table S3 (210 °C). The occurrence of internal diffusion limitation was determined according to the Weisz–Prater criterion (Eq. S7) [1, 2], where if C_{WP} (dimensionless parameter) is less than 1, internal diffusion limitation is absent.

r_A: reaction rate ($mol_{CO}\cdot kg_{Co}^{-1}\cdot s^{-1}$).

ρ_{Co}: Co density over the catalyst ($kg_{Co}\cdot m_{particle}^{-3}$).

R_c: average radius of the catalyst particle.

ε_p: porosity of the catalyst bed.

$D_{CO,\ wax}$: diffusion coefficient of CO in wax [3].

$H_{CO,wax}$: Henry coefficient, 2×10^{-2} $MPa\cdot m^3\cdot mol^{-1}$.

P_{CO}: CO partial pressure, 0.32 MPa.

Table SE1. The related parameters estimated by Weisz–Prater criterion

Catalyst	$-r_A$ ($\times 10^{-2} mol_{CO}\cdot kg_{Co}^{-1}\cdot s^{-1}$)	ρ_{Co} ($kg_{Co}\cdot m_{particle}^{-3}$)	R_c[c] ($\times 10^{-4}$ m)	$D_{CO,\ wax}$ (10^{-8} m^2 s^{-1})	C_{wp}
Co/SiO_2[a]	1.7	46.4	1.75	1.5	0.06
$Co\text{-}MnO_x/SiO_2$[a]	1.5	46.4	1.75	1.5	0.05
$Co\text{-}(C)/SiO_2$[a]	1.6	46.4	1.75	1.5	0.06
$Co\text{-}MnO_x\text{-}(C)/SiO_2$[a]	2.6	46.4	1.75	1.5	0.09
Co/SiO_2[b]	2.1	46.4	1.75	1.7	0.07
$Co\text{-}MnO_x/SiO_2$[b]	2.6	46.4	1.75	1.7	0.09
$Co\text{-}(C)/SiO_2$[b]	2.4	46.4	1.75	1.7	0.08
$Co\text{-}MnO_x\text{-}(C)/SiO_2$[b]	5.6	46.4	1.75	1.7	0.18

[a] The reaction data is acquired from Table S2 under the FTS reaction condition of 210 °C, 1.0 MPa.

[b] The reaction data is acquired from Table S3 under the FTS reaction condition of 220 °C, 1.0 MPa.

[c] The granular-sizes of catalysts (40-60 mesh) are about 350 μm, thus average radius of 175 μm are used to calculate

C_{wp}.

According to the calculated effectiveness factors listed in Table SE1, C_{WP} value is substantially less than 1 for the four tested catalysts under the reaction conditions of 210 ºC with CO conversion of 10 ± 1.4%, implying no diffusion limitations of the reactants on the surface reactions. Therefore, the measured intrinsic activity can be safely correlated with the chemical and structural properties of catalysts. When increasing reaction temperature to 220 ºC, the internal diffusion limitation still can be ignored in the cases of Co/SiO_2, $Co\text{-}MnO_x/SiO_2$, and $Co\text{-}(C)/SiO_2$ with CO conversion less than 35%. But for the catalyst of $Co\text{-}MnO_x\text{-}(C)/SiO_2$ with extremely high CO conversion of ~80% at 220 ºC, the relatively high C_{wp} value suggests some extent of the internal diffusion limitation on the rates of surface reactions.

In order to estimate the influence of external diffusion on the catalyst performance of $Co\text{-}MnO_x\text{-}(C)/SiO_2$ at 210 °C, the CO conversions were sequentially measured by increasing the reactant flow rate (volume of reactant per unit time) and catalyst volume (V) while keeping their ratio (space velocity, WHSV) constant [4]. The CO conversion and hydrocarbon product selectivity as function of flow rate were plotted and shown in Fig. SE8. Note that the flow rate has no influence on the CO conversion and product selectivity, as well as the chain growth probability, suggesting the external mass diffusion limitation is negligible for $Co\text{-}MnO_x\text{-}(C)/SiO_2$ with CO conversion less than 15% at 210 °C, 1.0 MPa.

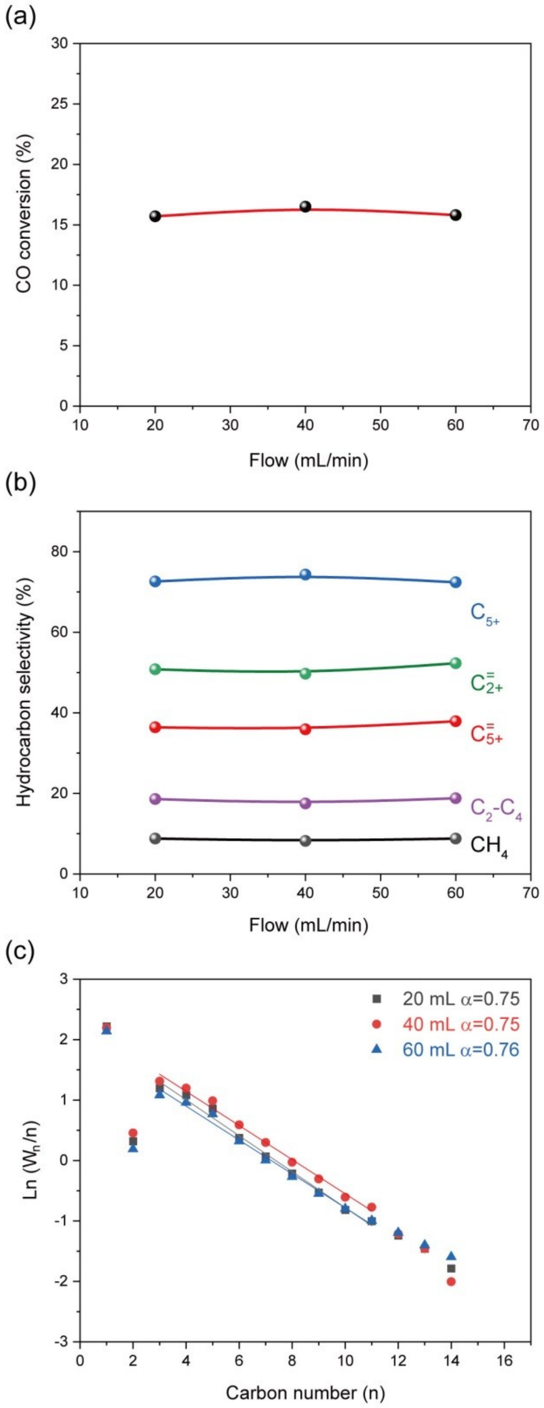

Fig. SE8. Flow rate test of Co-MnO_x-(C)/SiO_2 by keeping WHSV constant. The influences of flow rate on (**a**) CO conversion and (**b**) hydrocarbon product selectivity. (**c**) ASF plots with the corresponding chain growth probabilities in the C_3-C_{11} hydrocarbon range derived from online product analysis. Reaction conditions: 210 °C, 1.0 MPa, H_2/CO = 2, and the data is collected after more than 24 h on stream. Cat. = 0.25 g for 20 mL/min, 0.50 g for 40 mL/min, and 0.75 g for 60 mL/min.

2. Supplementary Figures

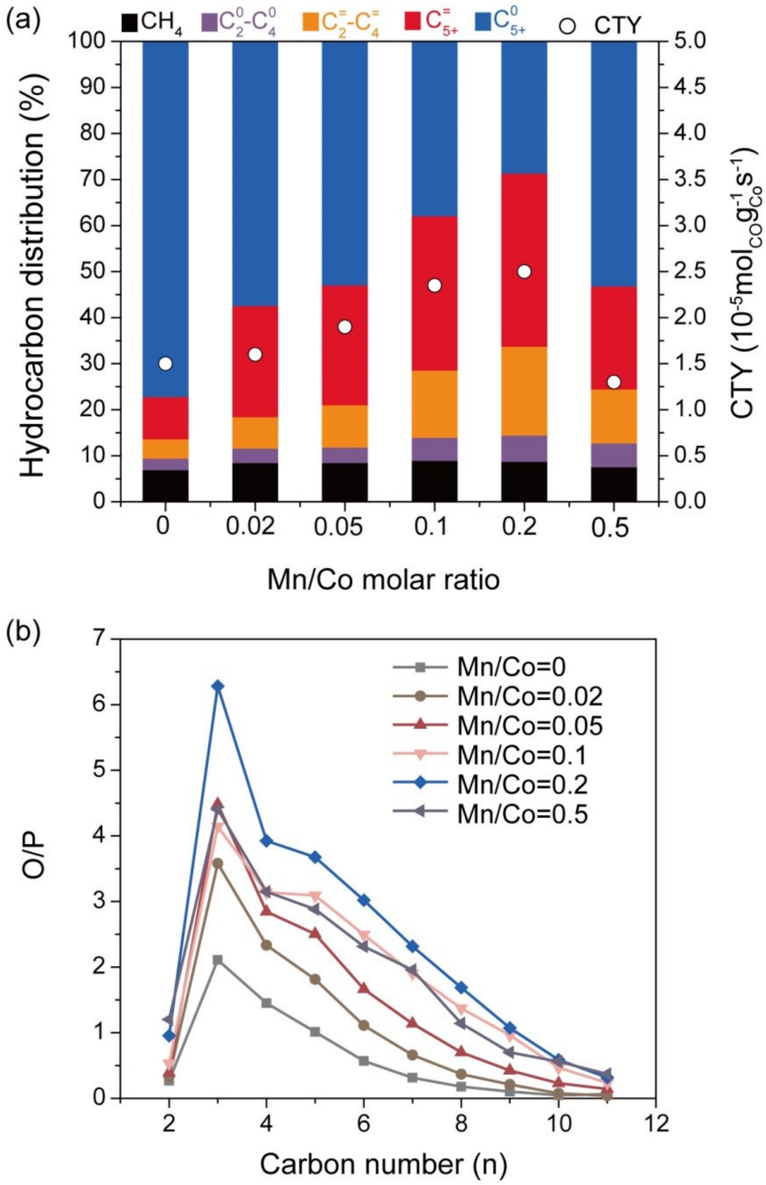

Fig. S1. FTS results over Co-MnO_x-(C)/SiO_2 with different Mn/Co molar ratios. (**a**) CTY and product distributions. (**b**) Olefin/paraffin (O/P) ratio of the products. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 210 °C, H_2/CO = 2.0, GHSV = 4.8 $L\cdot g_{cat}^{-1}\cdot h^{-1}$, and the data was collected after more than 24 h on stream.

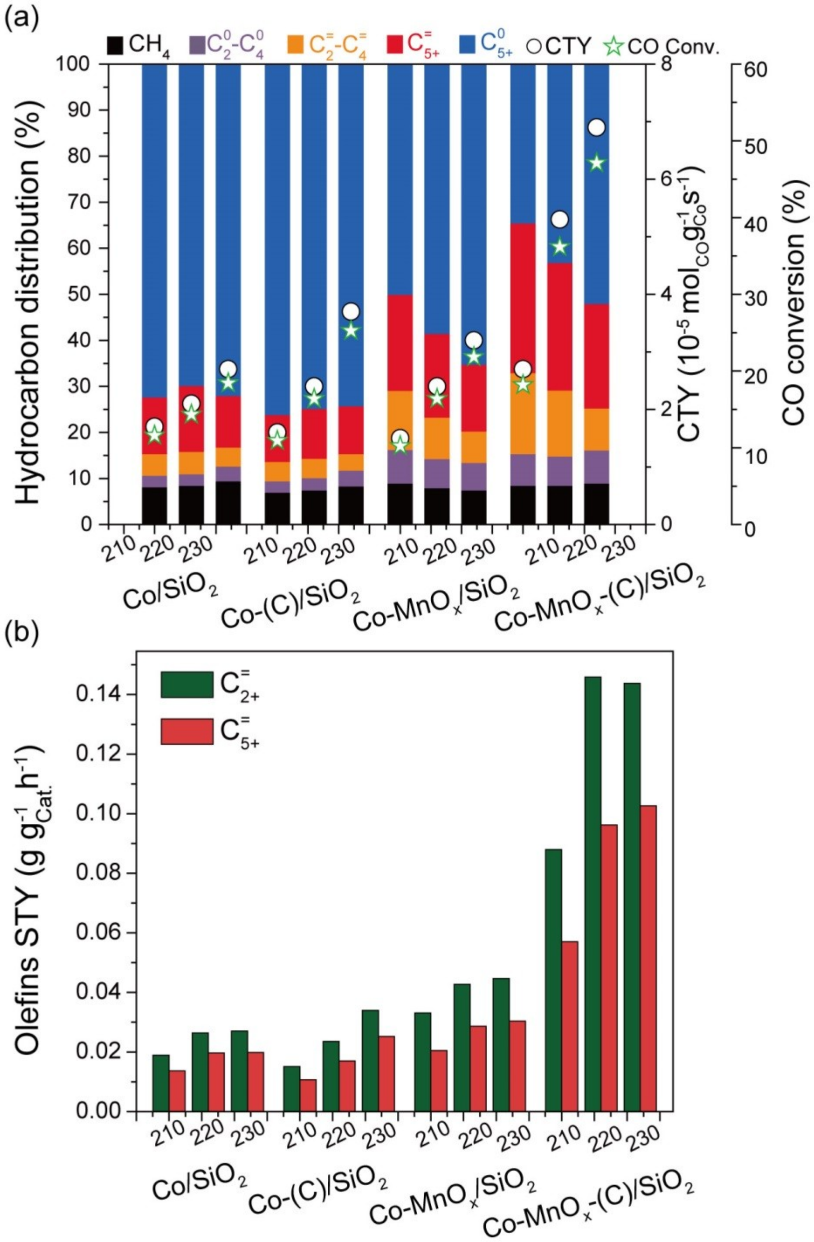

Fig. S2. FTS results of the four catalysts at different reaction temperatures. (**a**) CTY, CO conversion and product distributions. (**b**) Space-time yield (STY) of $C_{2+}^{=}$ and $C_{5+}^{=}$. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 4.8 $L \cdot g_{cat}^{-1} \cdot h^{-1}$, $H_2/CO = 2.0$, and the data was collected after more than 24 h on stream.

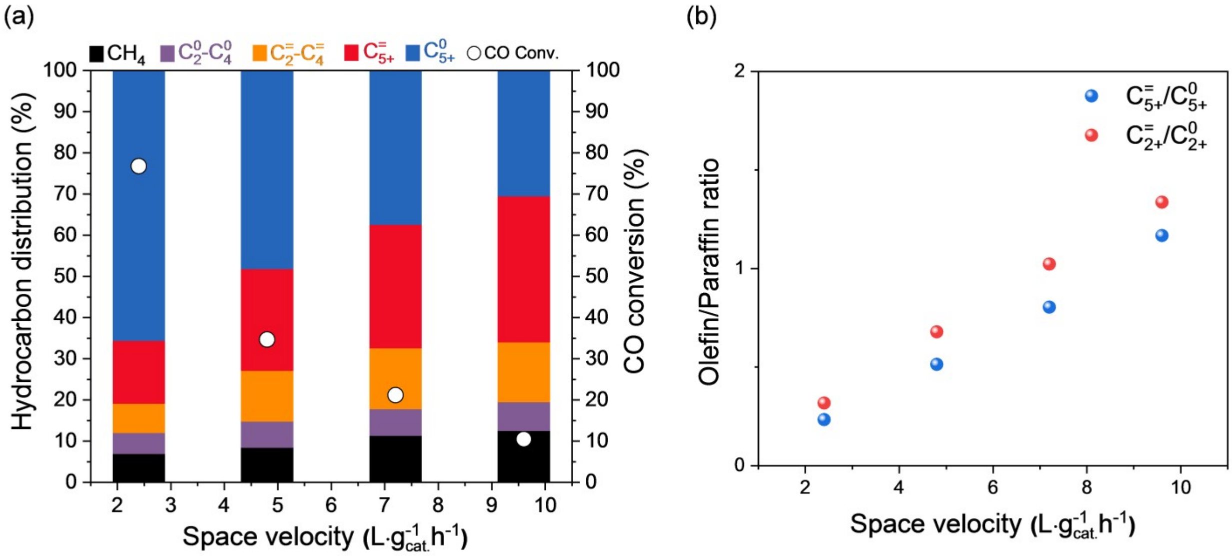

Fig. S3. (**a**) CO conversion and product distribution, and (**b**) O/P ratio of C_{2+} and C_{5+} over Co-MnO_x-(C)/SiO_2 at different GHSV. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 220 °C, GHSV = 2.4 ~ 9.6 $L\cdot g_{cat}^{-1}\cdot h^{-1}$, H_2/CO = 2.0, and the data was collected after 24 h on stream.

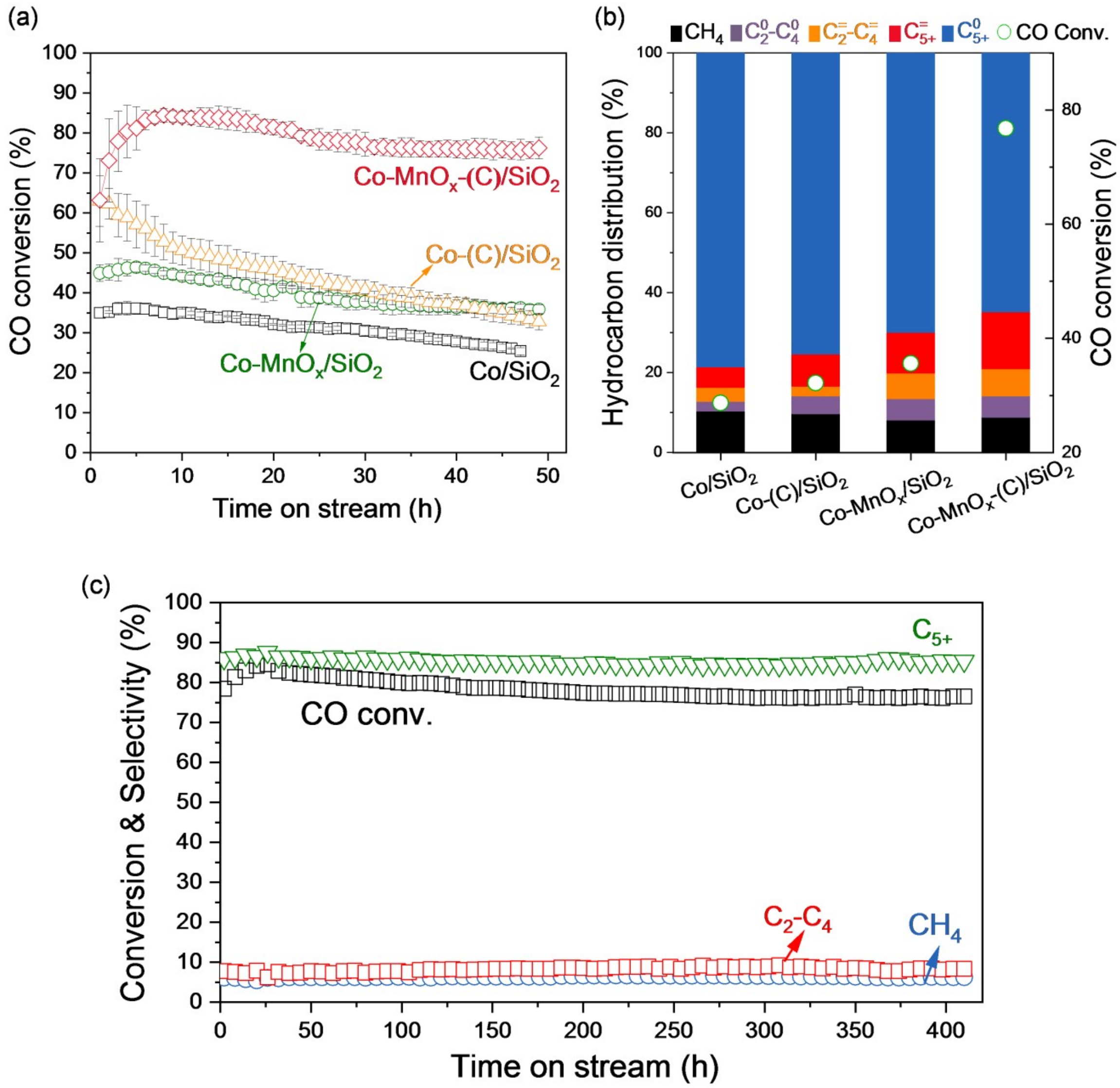

Fig. S4. Catalyst stability tests, (**a**) CO conversion as a function of the time-on-stream for the catalysts. (**b**) CO conversion and product distribution at the stable stage. (**c**) Long-term stability results over Co-MnO_x-(C)/SiO_2. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 220 °C, H_2/CO = 2.0, and GHSV = 2.4 $L \cdot g_{cat}^{-1} \cdot h^{-1}$.

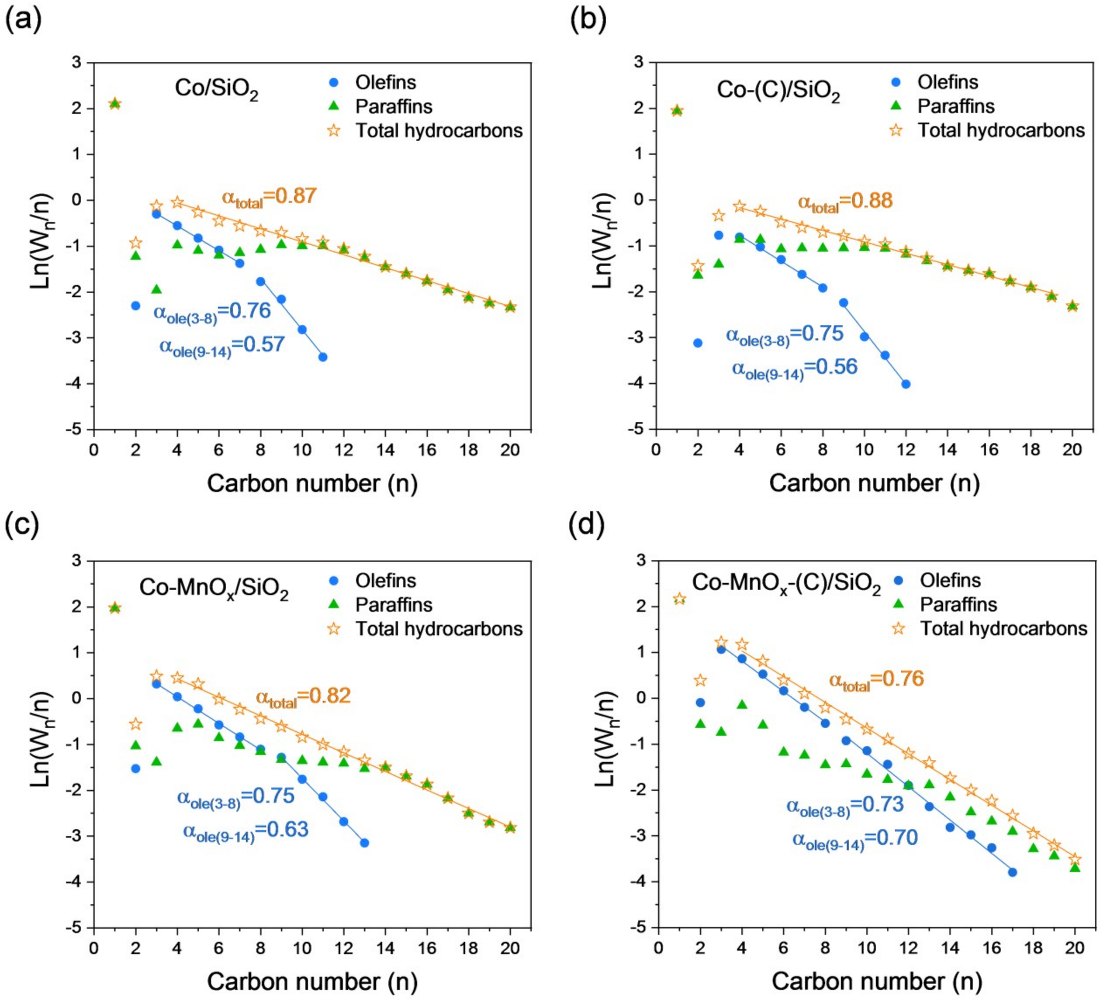

Fig. S5. ASF distribution of olefins and total hydrocarbons. (**a**) Co/SiO_2, (**b**) Co-(C)/SiO_2, (**c**) Co-MnO_x/SiO_2, and (**d**) Co-MnO_x-(C)/SiO_2. Reaction conditions: 0.50 g catalyst, 210 ºC, 1.0 MPa, H_2/CO = 2.0, GHSV of 4.8 to 7.2 L·g_{cat}^{-1}·h^{-1} for achieving the CO conversion at 10.0 ± 1.4%.

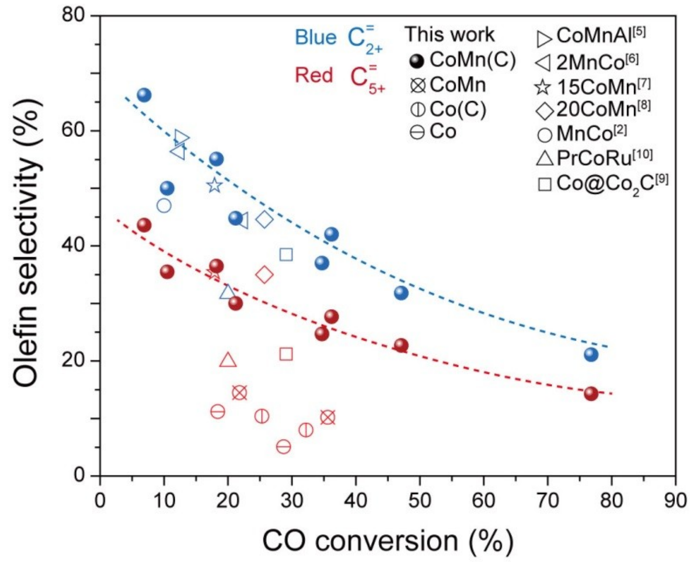

Fig. S6. The dependences of olefin selectivity on the CO conversion over Co-MnO_x-(C)/SiO_2 related catalysts and those reported in the references.

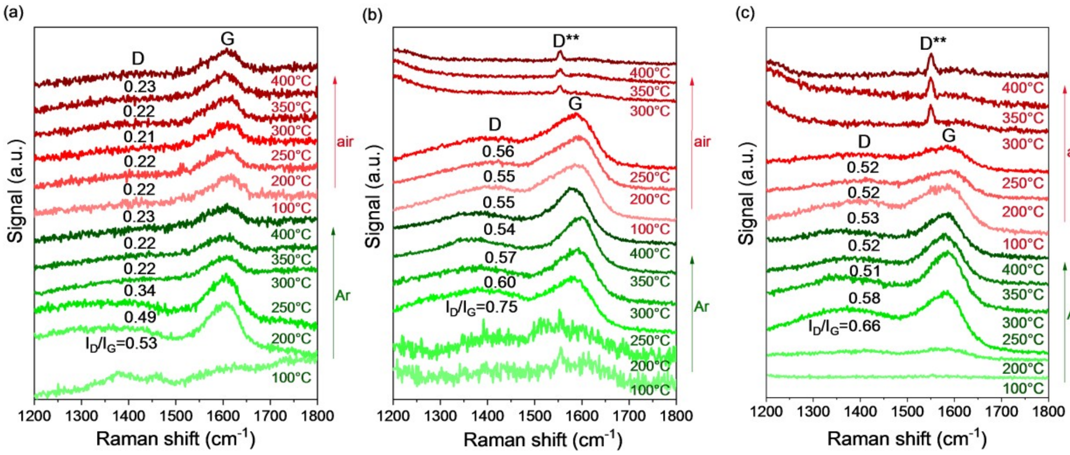

Fig. S7. *In situ* Raman spectra of the dried samples of (a) Glu/SiO_2, (b) Co-Glu/SiO_2, and (c) Co-Mn-Glu/SiO_2 during the consecutive treatments in the flow of Ar and air with increasing temperature.

In order to monitor the evolution of carbon species during heating process, the dried samples of SiO_2 impregnated with glucose (Glu/SiO_2), co-impregnation of $Co(NO_3)_2$ and glucose (Co-Glu/SiO_2), and co-impregnation of $Co(NO_3)_2$, $Mn(CH_3COO)_2$ and glucose (Co-Mn-Glu/SiO_2) were tested by *in situ* Raman technique during the consecutive treatments in a flow of Ar and air with increasing temperature. Specifically, 50 mg sample was loaded in the *in situ* cell (Harrick, HVC-DR2) and heated in a flow of Ar (20 mL/min) at atmospheric pressure with stepwise increasing temperature to 400 °C (5 °C/min). The corresponding spectrum at the 1200-1800 cm^{-1} region was recorded at 100, 200, 250, 300, 350 and 400 °C, respectively. After cooled down to room temperature, the sample was switched to a flow of air (20 mL/min) and heated again with stepwise increasing temperature to 400 °C (5 °C/min). The corresponding spectrum at the 1200-1800 cm^{-1} region was recorded at 100, 200, 250, 300, 350 and 400 °C, respectively.

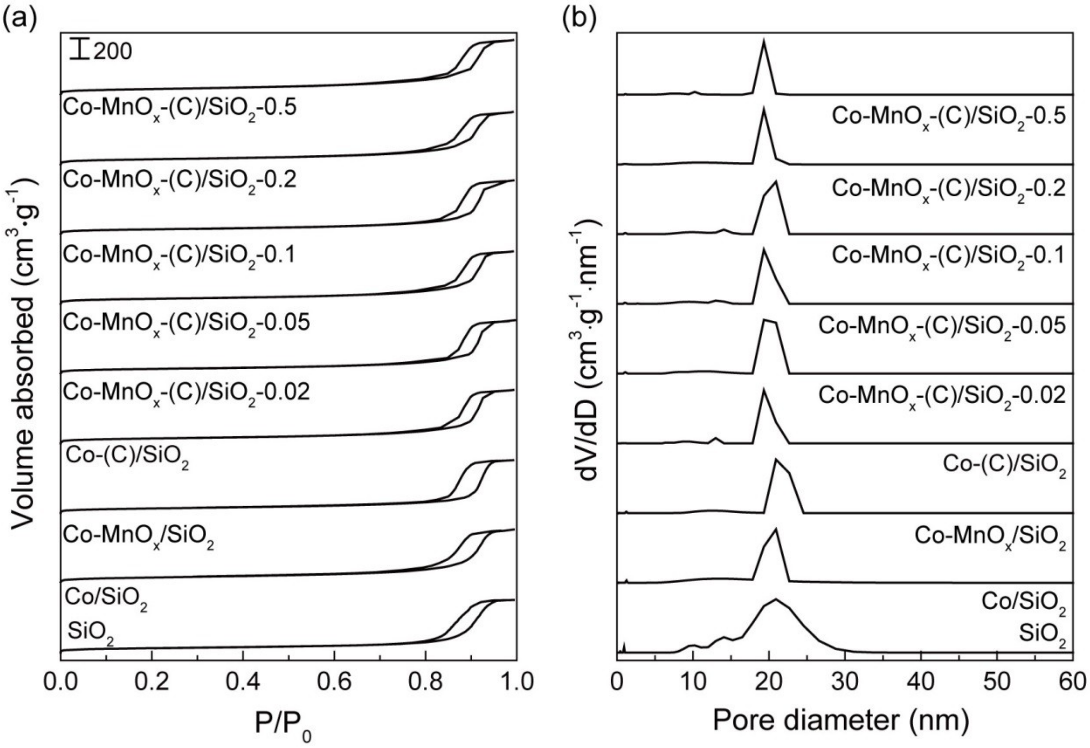

Fig. S8. Textural properties of the calcined catalysts. (**a**) N_2 adsorption-desorption isotherms. (**b**) Pore size distributions calculated by BJH mesopore model.

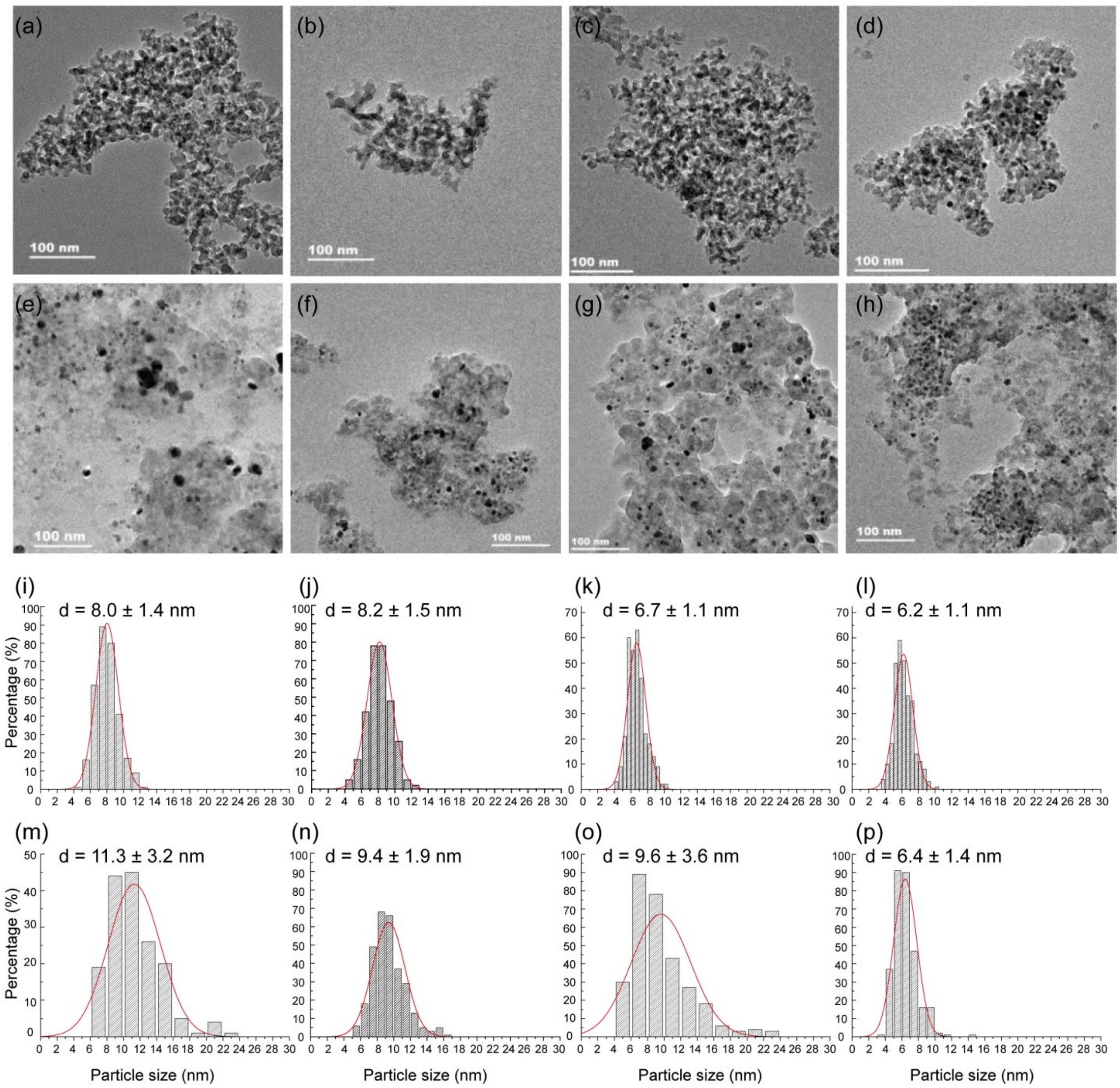

Fig. S9. TEM images, particle size distributions, and the average particle sizes of the calcined and used catalysts. Calcined catalysts: (**a, i**) Co/SiO_2, (**b, j**) Co-MnO_x/SiO_2, (**c, k**) Co-(C)/SiO_2, and (**d, l**) Co-MnO_x-(C)/SiO_2. Used catalysts: (**e, m**) Co/SiO_2, (**f, n**) Co-MnO_x/SiO_2, (**g, o**) Co-(C)/SiO_2, and (**h, p**) Co-MnO_x-(C)/SiO_2.

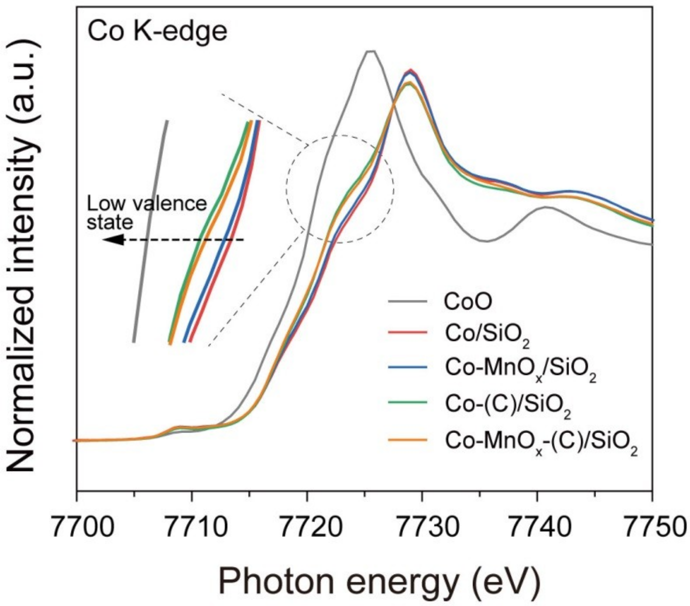

Fig. S10. Normalized XANES spectra of calcined catalysts at the Co K-edge with the standard CoO.

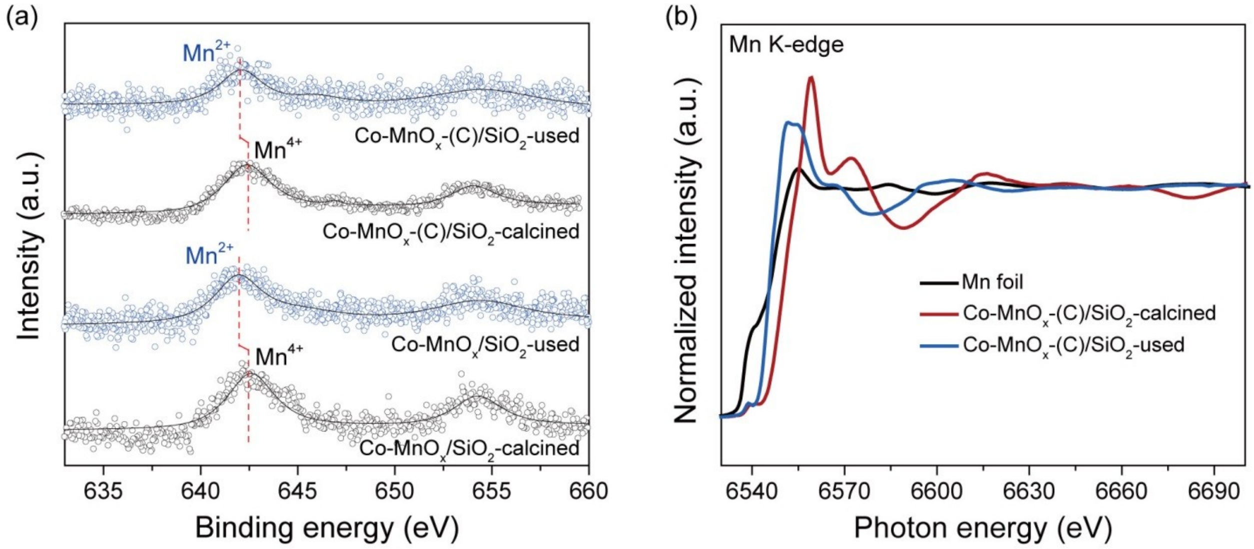

Fig. S11. (**a**) Mn 2p XPS spectra for the calcined and used catalysts. (**b**) Normalized Mn K-edge XANES spectra of the calcined and used Co-MnO_x-(C)/SiO_2 with the reference of Mn foil.

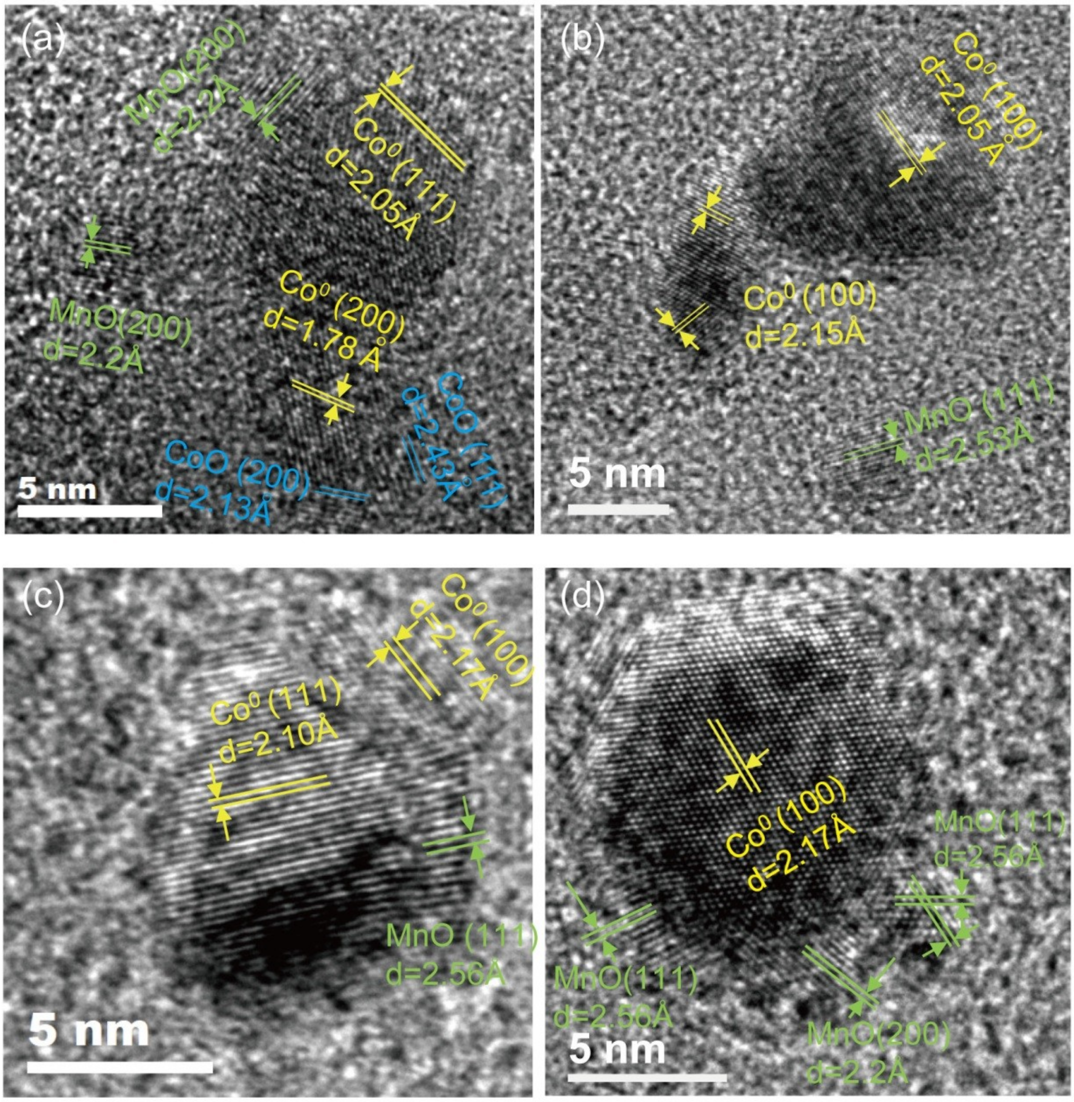

Fig. S12. HR-STEM images of the used catalysts. (**a-b**) Co-MnO_x/SiO_2, (**c-d**) Co-MnO_x-(C)/SiO_2. The spatial segregation of MnO_x and Co NPs is clearly seen on the used Co-MnO_x/SiO_2. In contrast, MnO_x is found in a close spatial proximity with Co NPs on the used Co-MnO_x-(C)/SiO_2.

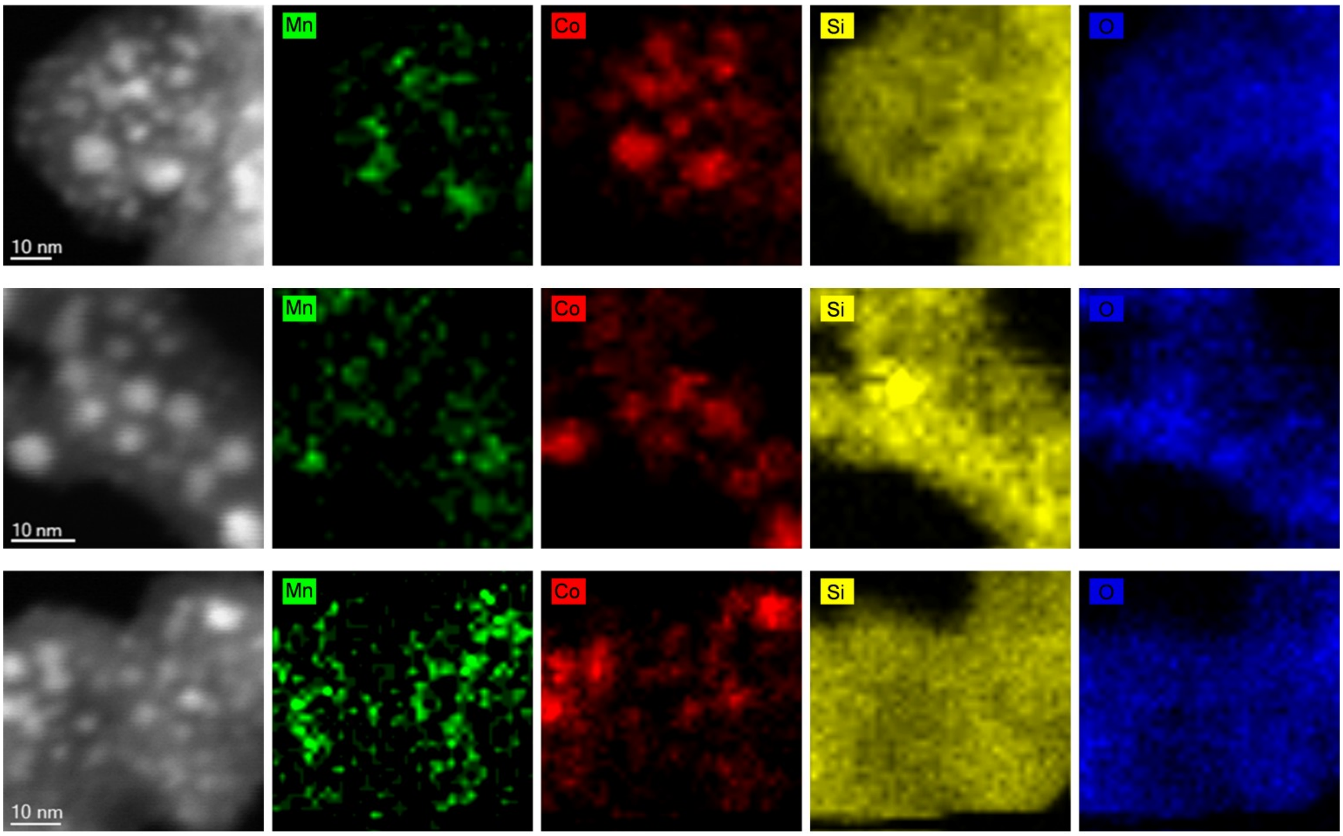

Fig. S13. High-magnification STEM-HAADF images on the random thin sample specimens of the used Co-MnO_x-(C)/SiO_2 and the corresponding EDS elemental maps of Mn (green), Co (red), Si (yellow), and O (blue).

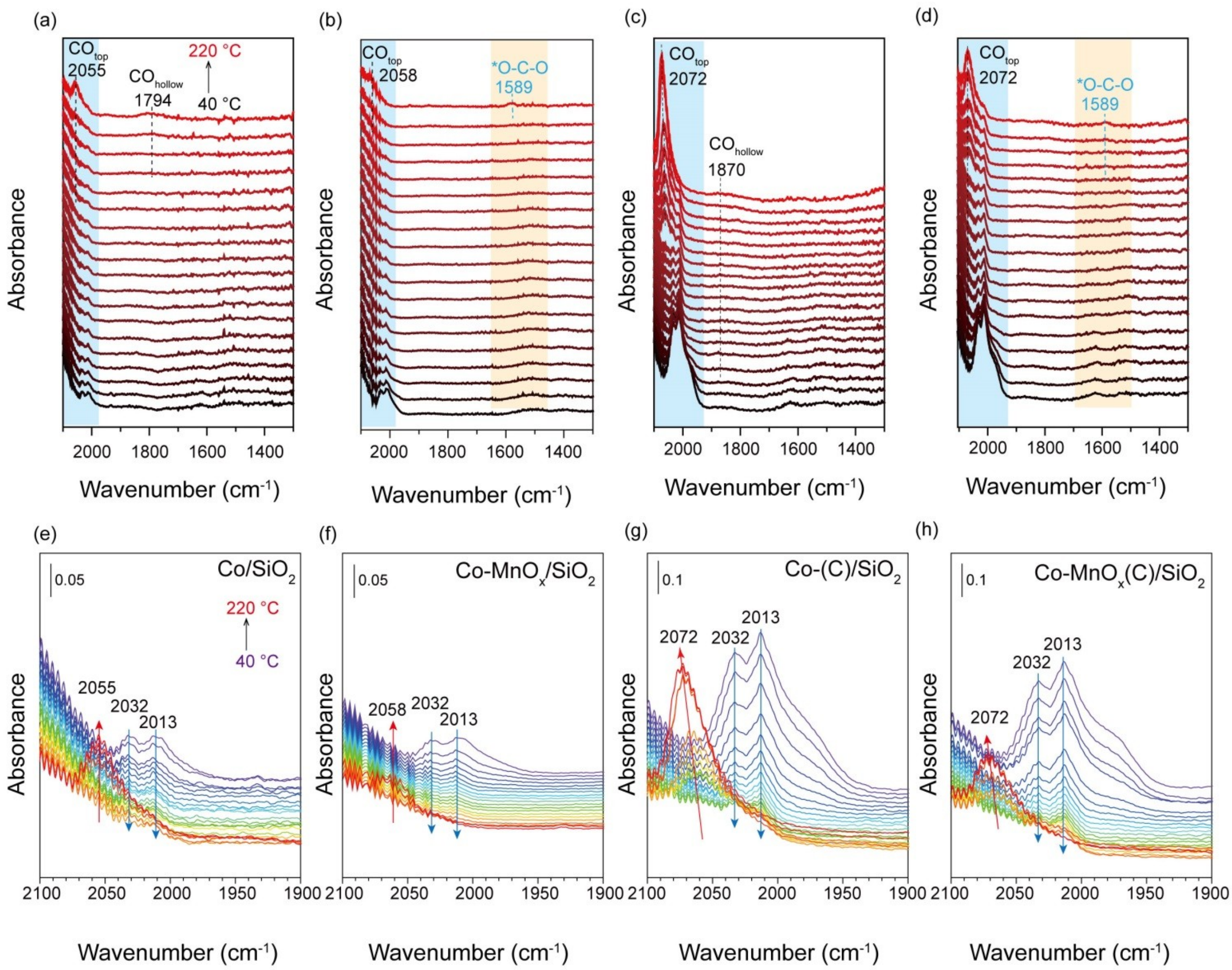

Fig. S14. *In situ* DRIFTS of CO as a function of temperature (40-220 °C) over the reduced catalysts: (**a**) Co/SiO_2, (**b**) Co-MnO_x/SiO_2, (**c**) Co-(C)/SiO_2 and (**d**) Co-MnO_x-(C)/SiO_2. Evolutions of CO_{top} with temperature in the wavenumber range of 2100-1900 cm^{-1}: (**e**) Co/SiO_2, (**f**) Co-MnO_x/SiO_2, (**g**) Co-(C)/SiO_2 and (**h**) Co-MnO_x-(C)/SiO_2. The spectra were collected from 40 to 220 °C under the conditions of 20 mL/min flow of 30%CO/Ar at the atmospheric pressure.

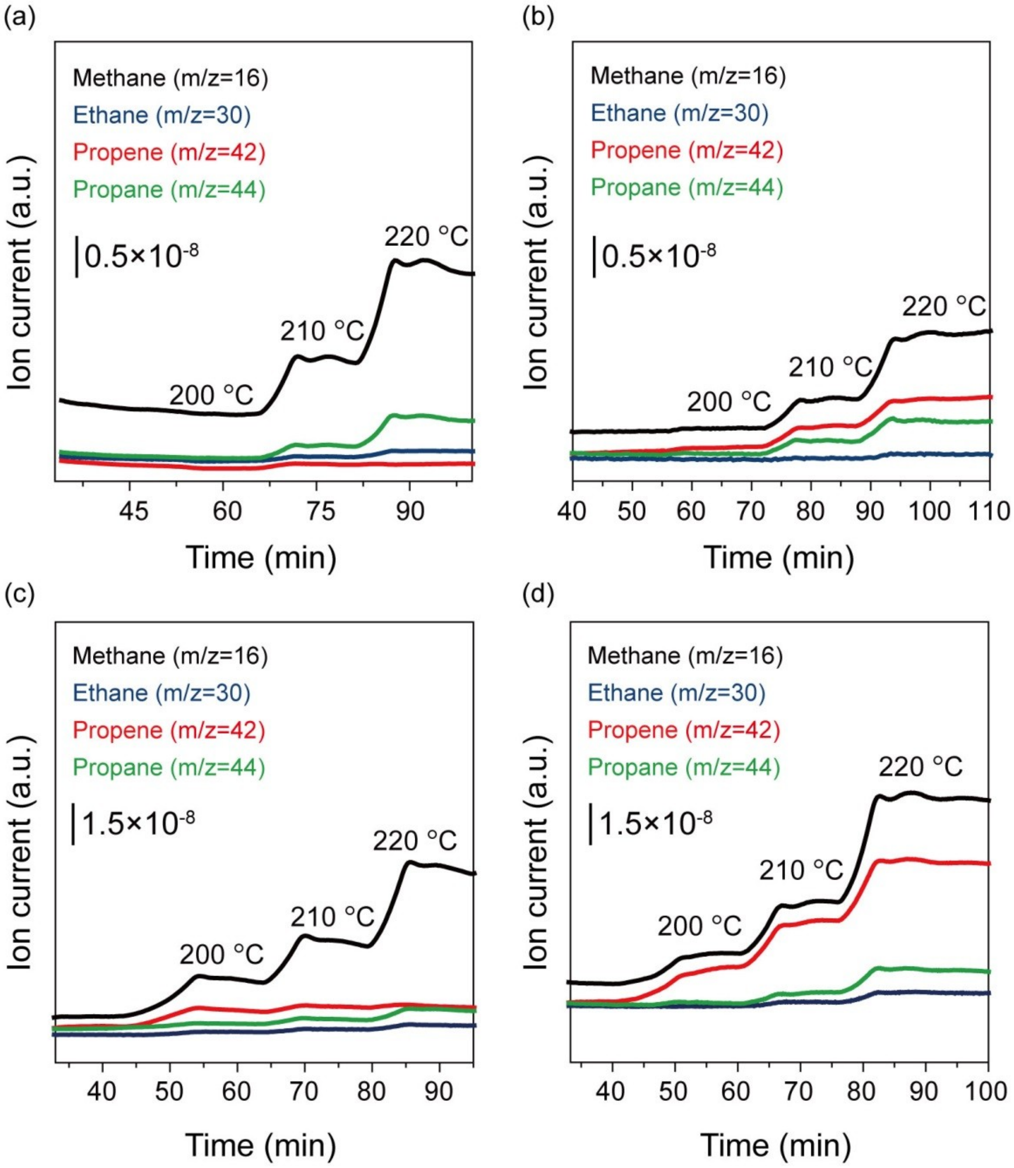

Fig. S15. The MS patterns of the gaseous hydrocarbons evolved during *in situ* DRIFTS of CO hydrogenation over (**a**) Co/SiO_2, (**b**) $Co\text{-}MnO_x/SiO_2$, (**c**) $Co\text{-}(C)/SiO_2$, and (**d**) $Co\text{-}MnO_x\text{-}(C)/SiO_2$.

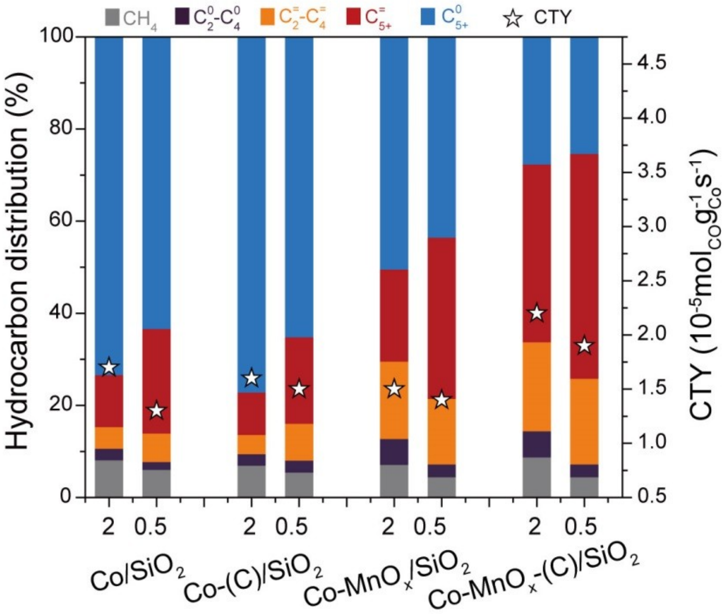

Fig. S16. Catalytic performances of the catalysts at different H_2/CO ratio of feed gas. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 4.8 $L\cdot g_{cat}^{-1}\cdot h^{-1}$. In order to achieve the similar CTY for each catalyst, the reaction temperature is adjusted, e.g., 210 ºC for H_2/CO = 2.0 and 220 ºC for H_2/CO = 0.5. The data was collected after more than 24 h on stream.

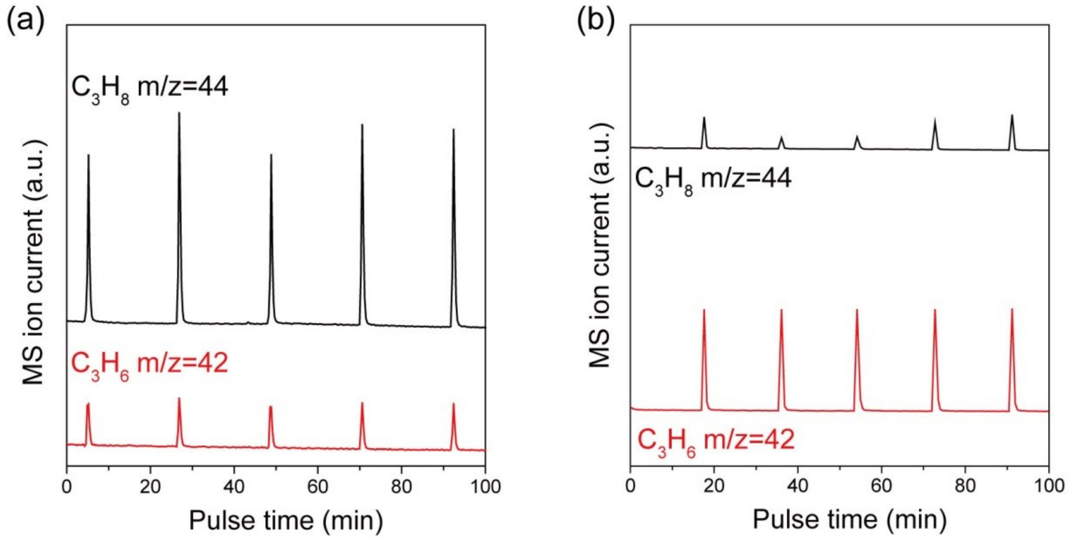

Fig. S17. Characterization of hydrogenation ability of (**a**) Co-(C)/SiO_2, (**b**) Co-MnO_x-(C)/SiO_2 on olefins by the experiment of pulsing propylene (20% C_3H_6 in Ar) into a flow of 10% H_2 in Ar. Before the pulse reaction, the catalysts (100 mg) were *in situ* reduced with 10% H_2 in Ar at 400 ºC for 2 h, then cooled down to 220 ºC. The C_3H_6 with m/z = 42 and C_3H_8 with m/z = 44 were detected by the mass spectrometry.

3. Supplementary Tables

Table S1. Metal loadings and the calculated textural properties

Catalyst	Co[a] (wt.%)	Mn[a] (wt.%)	Mn/Co molar ratio	Surface area[b] (m^2/g)	Pore volume[c] (cm^3/g)	Pore size[d] (nm)
SiO_2	n.a.[e]	n.a.[e]	n.a.[e]	223.4	1.11	19.9
Co/SiO_2	13.0	n.a.[e]	n.a.[e]	180.5	0.80	17.7
$Co-MnO_x/SiO_2$	12.9	2.5	0.21	211.8	1.06	19.7
$Co-(C)/SiO_2$	13.0	n.a.[e]	n.a.[e]	209.5	0.95	18.3
$Co-MnO_x-(C)/SiO_2-0.02$	13.1	0.2	0.02	203.3	1.01	19.8
$Co-MnO_x-(C)/SiO_2-0.05$	13.0	0.5	0.04	212.8	0.94	17.7
$Co-MnO_x-(C)/SiO_2-0.1$	12.6	1.0	0.09	198.4	0.97	19.4
$Co-MnO_x-(C)/SiO_2-0.2$	12.8	2.9	0.24	201.4	0.89	17.7
$Co-MnO_x-(C)/SiO_2-0.5$	12.3	5.6	0.49	221.1	0.93	16.7

[a] Measured by ICP-OES.

[b] Calculated by the Brunaue-Emmett-Teller (BET) method based on the adsorption branch of the N_2 adsorption/desorption isotherms.

[c] The total pore volume determined by the Barrett-Joyner-Halenda (BJH) method at a relative pressure of 0.99.

[d] The average pore diameter evaluated by the BJH method.

[e] n.a.: not available.

Table S2. Catalytic performances over prepared catalysts

Catalyst	Catalytic activity		Product selectivity (%)					
	Conversion (%)	CTY 10^{-5} $mol_{CO}\cdot g_{Co}^{-1}\cdot s^{-1}$	CH_4	CO_2	C_2-C_4	C_{5+}	$C_{2+}^{=}$	$C_{5+}^{=}$
Co/SiO_2	28.7	2.1	9.3	1.0	6.0	83.7	8.6	5.1
$Co\text{-}MnO_x/SiO_2$	35.6	2.6	7.2	0.9	11.8	80.1	16.7	10.2
$Co\text{-}(C)/SiO_2$	32.2	2.4	8.5	1.2	6.9	83.4	10.5	8.0
$Co\text{-}MnO_x\text{-}(C)/SiO_2$	76.8	5.6	7.0	1.8	12.1	79.1	21.1	14.3

Reaction conditions: 0.50 g catalyst, 220 ºC, 1.0 MPa, GHSV = 2.4 $L\cdot g_{cat}^{-1}\cdot h^{-1}$ and H_2/CO = 2.0. All the reaction data is collected after more than 24 h on stream.

Table S3. Catalytic performances over prepared catalysts

Catalyst	Catalytic activity			Hydrocarbon selectivity (%)					α[b]
	Conversion (%)	CTY $10^{-5}mol_{CO}\cdot g_{Co}^{-1}\cdot s^{-1}$	TOF[a] ($10^{-2}s^{-1}$)	CH_4	C_2-C_4	C_{5+}	$C_{2+}^{=}$	$C_{5+}^{=}$	
Co/SiO_2	11.4	1.7	3.3	8.2	7.2	84.6	16.1	11.3	0.87
Co-MnO_x/SiO_2	9.1	1.5	3.6	7.2	22.3	70.5	35.8	19.0	0.82
Co-$(C)/SiO_2$	11.2	1.6	3.9	7.0	6.8	86.2	13.4	9.2	0.88
Co-MnO_x-$(C)/SiO_2$	11.5	2.6	6.4	8.8	25.8	65.4	55.8	36.9	0.76

Reaction conditions: 0.50 g catalyst, 210 °C, 1.0 MPa, GHSV = 4.8~7.2 $L\cdot g_{cat}^{-1}\cdot h^{-1}$, H_2/CO = 2.0. All the reaction data is collected after more than 24 h on stream. The CO_2 selectivity is ignored as it is less than 1% for each tested catalyst.

[a] The turnover frequency (TOF) was calculated based on the reaction results at an initial TOS of 1-3 h.

[b] The carbon-chain growth probability of α is calculated based on C_3-C_{10} products.

Table S4. Comparison of catalytic performances with the typical Co-based FTS catalysts published in references for long-chain olefins

Catalyst	H_2/CO	Temp. (°C)	Pressure (MPa)	GHSV[a]	CO Conv. (%)	CTY[b]	Selectivity (%) CH_4	Selectivity (%) CO_2	Selectivity (%) $C_{2+}^{=}$	Selectivity (%) $C_{5+}^{=}$	STY (g/g_{Cat}/h) $C_{5+}^{=}$	STY (g/g_{Co}/h) $C_{5+}^{=}$
1.5Mn10Co-SiO_2[2]	2.0	220	2.0	n.a.	10~15	1.6	11.0	<1	47.0	n.a.	n.a.	n.a.
CoMnAl@SiO_2[5]	0.5	260	1.0	4.0	12.5	n.a.	6.5	15.1	58.8	n.a.	n.a.	n.a.
2Mn-Co@C[6]	0.5	250	0.5	2.0	12.4	0.4	12.8	5.8	56.4	n.a.	n.a.	n.a.
2Mn-Co@C[6]	1.0	250	0.5	2.0	22.3	0.7	16.2	4.2	44.4	n.a.	n.a.	n.a.
15CoMn/SiO_2[7]	2.0	230	1.0	4.5	17.9	2.1	7.7	1.0	50.5	35.4	0.056	0.37
20Co5Mn/SiO_2[8]	2.0	230	1.0	4.5	25.7	2.3	7.1	0.5	44.6	35.0	0.080	0.40
Co@Co_2C[9]	2.0	220	3.0	2000[c]	29.1	2.5	8.1	2.4	38.5	21.2	0.040	0.26
1.0PrCoRu/Al_2O_3[10]	2.0	200	2.0	n.a.	20±3	1.0	8.4	0.9	31.7	19.9	0.015	0.10
Co-MnO_x-(C)/SiO_2 (This work)	2.0	200	1.0	4.8	6.9	1.0	8.9	0.1	22.6	43.6	0.029	0.22
	2.0	210	1.0	4.8	18.6	2.7	8.5	0.5	50.1	32.5	0.057	0.45
	2.0	220	1.0	4.8	36.2	5.3	8.5	0.5	42.0	27.7	0.096	0.74
	2.0	230	1.0	4.8	47.1	6.9	9.0	1.0	31.8	22.7	0.103	0.79

[a] GHSV, $L\cdot g_{cat}^{-1}\cdot h^{-1}$
[b] CTY, 10^{-5} $mol_{CO}\cdot g_{Co}^{-1}\cdot s^{-1}$.
[c] GHSV of 2000 h^{-1} is used.
n.a.: not available.

Table S5. Summary of the binding energies for different forms of Co from the deconvoluted XPS spectra

Catalyst	Calcined catalyst			Reduced catalyst[a]			Used catalyst[b]		
	Co $2p_{3/2}$ (eV)		Atomic ratios	Co $2p_{3/2}$ (eV)		Atomic ratios	Co $2p_{3/2}$ (eV)		Atomic ratios
	Co^{3+}	Co^{2+}	Co^{2+}/Co^{3+}	Co^{2+}	Co^{0}	$Co^{0}/(Co^{0}+Co^{2+})$	Co^{2+}	Co^{0}	$Co^{0}/(Co^{0}+Co^{2+})$
Co/SiO_2	779.1	780.8	1.8	780.8	778.4	0.36	780.8	778.3	0.52
$Co\text{-}MnO_x/SiO_2$	779.0	780.6	2.4	781.4	778.8	0.35	781.4	778.3	0.77
$Co\text{-}(C)/SiO_2$	779.2	780.9	7.4	782.2	778.2	0.04	781.1	778.3	0.25
$Co\text{-}MnO_x\text{-}(C)/SiO_2$	779.4	781.0	7.9	782.6	778.4	0.03	781.5	778.6	0.51

[a] After reduction in 10%H_2 in Ar (10 mL/min) at 400 ºC for 3 h.

[b] After reaction in syngas flow (32%CO/64%H_2/4%Ar, 10 mL/min) at 220 ºC for 10 h.

Table S6. Co K-edge EXAFS fitting results for calcined samples and the reference Co_3O_4

Sample	**Scattering pair**	**CN**	**R(Å)**	**σ^2 (10^{-3}Å^2)**	**ΔE_0(eV)**	**R-factor**
Co_3O_4	Co-O1	6	1.91±0.01	1.2±0.9	-0.97±1.4	0.0028
	Co-Co1	6	2.86±0.01	4.0±0.8	2.38±1.0	
	Co-Co2	6	3.35±0.02	4.3±1.3	2.38±1.0	
	Co-O1	4	1.82±0.04	4.0±0.8	-0.97±1.4	
	Co-Co1	12	3.35±0.03	4.0±0.8	2.38±1.0	
Co/SiO_2	Co-O1	4.6±1.2	1.93±0.01	1.3±1.8	3.17±1.7	0.0012
	Co-Co1	5.5±1.5	2.86±0.01	3.4±1.5	3.17±1.7	
	Co-Co2	5.5±1.5	3.36±0.01	4.4±0.9	3.17±1.7	
	Co-O1	2.8±1.1	1.86±0.03	3.4±1.5	3.17±1.7	
	Co-Co1	2.8±1.1	3.29±0.05	3.4±1.5	3.17±1.7	
Co-MnO_x/SiO_2	Co-O1	4.4±0.9	1.91±0.03	2.4±1.3	4.03±1.2	0.0009
	Co-Co1	5.6±1.1	2.87±0.01	4.0±1.3	4.03±1.2	
	Co-Co2	5.6±1.1	3.37±0.01	4.5±0.6	4.03±1.2	
	Co-O1	2.5±0.8	1.93±0.09	4.0±1.3	4.03±1.2	
	Co-Co1	2.5±0.8	3.33±0.05	4.0±1.3	4.03±1.2	
Co-(C)/SiO_2	Co-O1	4.0±0.6	1.90±0.06	3.8±6.2	1.54±1.3	0.0006
	Co-Co1	3.2±0.8	2.84±0.01	2.5±1.4	1.54±1.3	
	Co-Co2	3.2±0.8	3.36±0.01	3.9±0.9	1.54±1.3	
	Co-O1	2.2±0.6	1.94±0.01	2.5±1.4	1.54±1.3	
	Co-Co1	2.2±0.6	3.26±0.12	2.5±1.4	1.54±1.3	
Co-MnO_x-(C)/SiO_2	Co-O1	4.1±0.6	1.90±0.01	2.7±1.2	2.66±1.0	0.0004
	Co-Co1	4.4±0.8	2.86±0.01	4.4±1.1	2.66±1.0	
	Co-Co2	4.4±0.8	3.37±0.01	5.3±0.5	2.66±1.0	
	Co-O1	2.3±0.5	1.95±0.04	4.4±1.1	2.66±1.0	
	Co-Co1	2.3±0.5	3.31±0.03	4.4±1.1	2.66±1.0	

CN, coordination number; *R*, interatomic distance; *σ*, disorder parameter; ΔE_0, energy shift. All the fitting analysis were performed in the *R* space, ΔR = 1.0-3.4 and ΔK = 3.2-12.8. S_0^2 = 0.59 was obtained from fitting the Co_3O_4 standard sample.

Table S7. The calculated reduction degree, H_2 uptake and particle size of different catalysts

Catalyst	Reduction degree[a] (%)	H_2-chemisorption			TEM	XRD	
		H_2 uptake[b] ($\mu mol \cdot g_{cat}^{-1}$)	Dispersion[c] (%)	Particle size[d] (nm)	Particle size[e] (nm)	Co_3O_4 size[f] (nm)	Co^0 size[g] (nm)
Co/SiO_2	85.7	32.9	6.9	13.8	9.0	17.8	13.4
$Co\text{-}MnO_x/SiO_2$	71.5	30.5	7.8	12.3	8.2	12.4	9.3
$Co\text{-}(C)/SiO_2$	59.5	41.2	12.6	7.6	6.7	6.9	5.2
$Co\text{-}MnO_x\text{-}(C)/SiO_2$	42.8	25.6	11.3	8.5	6.2	6.3	4.7

[a] Determined by the O_2 titration method.

[b] Calculated based on the H_2-chemisorption.

[c] Dispersion percentage calculated from $D = 1.179X/(W \cdot f)$, where X-H_2 uptake, f-reduction degree and W-the weight percentage of cobalt.

[d] Determined by H_2-chemisorption and calibrated by reduction degree.

[e] Determined from TEM images of calcined catalysts and calculated upon the measurement of random 300 particles.

[f] Calculated by Scherrer equation applied to the most intense diffraction ($2\theta = 36.8°$) from XRD results of the calcined catalysts.

[g] Calculated according to $d(Co^0) = 0.75d(Co_3O_4)$.

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